TEMPLATE

KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

PUBLICATION DATE **29.06.2023**

VERSION **v.1.5**

RELATED SUPPORT

- TEMPLATE GUIDE Key Project Information & Project Design Document

This document contains the following Sections

This template has been revised to aid a consistent interpretation and to better support project developers submitting documentation for certification. Please read the accompanying guide to understand how to complete this template accurately.
TEMPLATE GUIDE Key Project Information & Project Design Document

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KEY PROJECT INFORMATION

GS ID of Project	11352
Title of Project	Kenya biomass gasification for clean cooking
Time of First Submission Date	06/09/2021
Date of Design Certification	29/09/2023
Version number of the PDD	4.0
Completion date of version	10/10/2023
Project Developer	BetterCooking Company Limited
Project Representative	Tom Price
Project Participants and any communities involved	BetterCooking Company Limited
Host Country (ies)	Kenya
Activity Requirements applied	☒ Community Service Activity ☐ Renewable Energy ☐ Land-Use and Forests Activity Requirements/Risks & Capacities ☐ N/A
Scale of the project activity	☒ Micro scale ☐ Small Scale ☐ Large Scale
Other Requirements applied	Microscale project requirements, CSA requirements

Methodology (ies) applied and version number	Gold Standard "Methodology for metered & measured energy cooking devices"[1], version 1.20
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☒ Regular ☐ Retroactive

Table 1 – Estimated Sustainable Development Contributions

SUSTAINABLE DEVELOPMENT GOALS TARGETED	SDG IMPACT (DEFINED IN B.6)	ESTIMATED ANNUAL AVERAGE	UNITS OR PRODUCTS
13 Climate Action (mandatory)	Reduced GHG emissions	3,823	t CO2e
3 Good health and wellbeing	- Avoided kitchen smoke exposure - Reduced health impact	-420 -1,360	person-years person-years
5 Gender Equality	Time saved by women due to faster cooking	93,075	hours saved per year in total
7 Affordable and clean energy	Access to clean cooking technology	6,630	person-years
8 Decent work and economic growth	Employment generation	25	full-time jobs

[1] https://globalgoals.goldstandard.org/430_ee_ics_methodology-for-metered-measured-energy-cooking-devices/

SECTION A. DESCRIPTION OF PROJECT

A.1 Purpose and general description of project

The project replaces traditional cooking methods in Kenya with a focus on Nairobi. The technology that will be employed is EcoSafi's clean cooking system, utilizing gasification cookstoves and clean burning pellets from renewable/residual biomass. The project boundary includes the locations of cooking as well as baseline and project fuel collection/production. In the baseline scenario, households and institutions (e.g. small restaurants) use non renewable biomass (mainly charcoal) and partially fossil fuels for cooking, leading to GHG emissions and negative health effects, consequences.

Clean cookstoves are distributed directly by project staff. All users sign contracts where they confirm that carbon rights are ceded.

It is planned to distribute > 2,000 gasification cookstoves under this project. As of October 2022, about 1,200 clean cookstoves have been distributed in Nairobi.

Below is a rough indicative timeline of the project:

June 5 2021	Local stakeholder meeting in Nairobi
Q2 2021	First pilot distribution of stoves in Donholm, Nairobi (These stoves are not considered under the project)
from June 16 2021 on	Distribution of >2,000 stoves in Nairobi, set -up of different pellet distribution hubs

A.1.1. Eligibility of the project under Gold Standard

Eligibility criteria as per section 3.1.1 of GS4GG Principles & Requirements

(a) Type of Project

As per section 4.1.3 of GS4GG Principles & Requirements, v1.2, a project type is automatically eligible for Gold Standard Certification if there are approved Gold Standard Activity Requirements and/or Gold Standard Impact Quantification Methodologies associated with it or as referenced in Gold Standard Product Requirements.

Since the project will introduce clean cooking to households and institutions, it falls under project type category b) as defined in the GS Community service Activity requirements[2]: End-use energy efficiency.
As a result, the Project is automatically eligible for Gold Standard certification.

(b) Location of Project

The project is located within the host country i.e., Kenya. This is eligible as per the GS4GG eligibility requirements, as a project can be located in any part of the world.

(c) Project area, Project Boundary & Scale

Project Area & Boundary

The project boundary comprises the physical, geographical sites of the project technology (clean cookstoves) and baseline and project fuel collection and production, in accordance with the Gold Standard "Methodology for metered & measured energy cooking devices"[3], version 1.2.

Project Scale

The project is a micro scale project, generating fewer than the threshold of 10,000 tCO_2 emission reductions annually.

Double counting

Devices distributed under the project will not be included in any other carbon project or program. Double counting of emission reductions is avoided through the unique identification number assigned to of each customer. Each clean cookstove's serial number is also registered (and linked to the relevant unique customer identification number).. In addition, mobile money transaction numbers are used as evidence for pellets sales which are the basis for emission reduction generation[4]. As a result of this

[2] Community Services Activity Requirements – Gold Standard for the Global Goals

[3] https://globalgoals.goldstandard.org/430_ee_ics_methodology-for-metered-measured-energy-cooking-devices/

[4] In case customers return faulty pellets (e.g. wet, dusty, etc.) they are replaced with new pellets, which are then used. To date, no EcoSafi customers have returned unused pellets. Even when customers have returned stoves and cancelled their pellet subscriptions, they have used all pellets prior to doing so. EcoSafi expects this to continue.

three-tiered system of tracking (customer, stove, mobile money transaction linked to pellet sales), double counting impossible will no occur.

(d) Host Country Requirements

No legal, environmental, ecological and social regulations in Kenya prevent the distribution of clean cookstoves to households and institutions (e.g. small restaurants). Hence, the project is in compliance with the host country's legal, environmental, ecological and social regulations.

(e) Contact Details

The project is managed by BetterCooking Company Limited, registered in the USA; and by Ecosafi, a Kenyan company owned by BetterCooking.

Name	BetterCooking Company Limited
Contact Details	102 Quarry Lane, PO Box 1908, Karen, Nairobi, 00502
Legal Registration Details	Kenyan company registration number PVT-EYUJPVK
Justification on the Entity is in good standing	No insolvency or legal/criminal notices filed against Better Cooking Company Limited or any of its Directors.

(f) Legal Ownership

All project devices (Clean Cook Stoves) will be provided by EcoSafi under a leasing model with a small, refundable deposit to the end user.The expected GS VERs and other potential environmental benefit entitlements will held by EcoSafi, because EcoSafi will sign an agreement with each customer to transfer all carbon rights and other environmental benefits to EcoSafi upon becoming a customer. Additionally, pellet providers will sign agreements on ceding carbon rights to EcoSafi.

(g) Other Rights

(h) Stoves are owned by EcoSafi and provided to customers for use per the signed customer agreement. Each EcoSafi customer waives their rights to the emission reductions and all other environmental benefits that result from stove use per the signed customer agreement.**ODA Declaration**

The project will not make use of public funding from Annex I parties that could result in a diversion of official development assistance. There is no emission reduction cap or emissions trading programme in Kenya[5] that could get in conflict with the project. Double counting with similar carbon projects is prevented by unique identification of stove users (see A.1.1 C) above).

Eligibility criteria as per GS4GG Community Services Activity Requirements

No.	Eligibility Criterion	Demonstration of eligibility of the project
2.1.2	CSA Projects shall lead to climate change mitigation and/or adaptation by providing or improving access to services/resources at household or community or institution level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.	The project distributes clean cookstoves to households and institutions (e.g. small restaurants), therefore providing (clean) energy services.
2.1.3	In relation to the above all Projects shall therefore conform to Gold Standard for the Global Goals Principles & Requirements (and associated documents).	Conformity with the GS4GG Principles & Requirements is discussed above.
3.1.1 (b)	**Types of project** – Pre-identified CSA project types are noted below. Project Developers may submit new project types to Gold Standard for approval following the Principles & Requirements. (a) … **b) End- Use Energy efficiency:** Project activities that reduce energy requirements as compared to baseline	See previous criterion No. 2.1.2 above, showing that End-Use Energy Efficiency activities are implemented

[5] https://climatechampions.unfccc.int/africa-carbon-markets-initiative-announces-13-action-programs/ The set up of the mentioned programmes shows that the voluntary market is legal in Kenya).

	scenario without affecting the level and quality of services or products, where the end user of the products and services are clearly identified and when the physical intervention is required at the user end. For example, efficient cooking, heating, lighting, etc.	
3.1.2 (a)	**Project Area, Boundary and Scale:** Project Area and Boundary shall be defined in line with the applicable Methodologies or Product Requirements. The definition of scale is the same for all Projects, except Microscale which is defined as: (a) Project issuing emission reductions less than or equal to 10,000 tCO2eq per annuum	The project boundary comprises the physical, geographical sites of the project technology (clean cookstoves) and baseline and project fuel collection and production, in accordance with the Gold Standard "Methodology for metered & measured energy cooking devices", version 1.2. The annual emission reductions are capped at 10,000tCO2eq per annuum
3.1.4 a)	**Legal ownership:** Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided.	See equivalent criterion under GS4GG Principles & Requirements above.
3.1.4 b)	The transfer of Product ownership shall be discussed during local stakeholder consultations for regular cycle projects. For retroactive projects, the project participants shall collect stakeholder feedback through live consultations, telephone discussions, electronic mode, etc. as deemed necessary to reach out to the relevant stakeholders.	The transfer of product ownership was discussed at the Local Stakeholder Consultation (LSC).

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

EcoSafi is the legal owner of the emission reductions. The same was discussed and agreed on during the LSC meeting. Ecosafi signs an agreement with each customer confirming the transfer of all legal rights of the emission reductions and all other environmental benefits. Additionally, pellet providers sign agreements that cede rights to to the emission reductions and all other environmental benefits.

A.2 Location of project

Republic of Kenya

The project area involves the entire country. Initial activities are taking place in Nairobi.

Coordinates of the capital Nairobi: 1°16′S 36°48′E

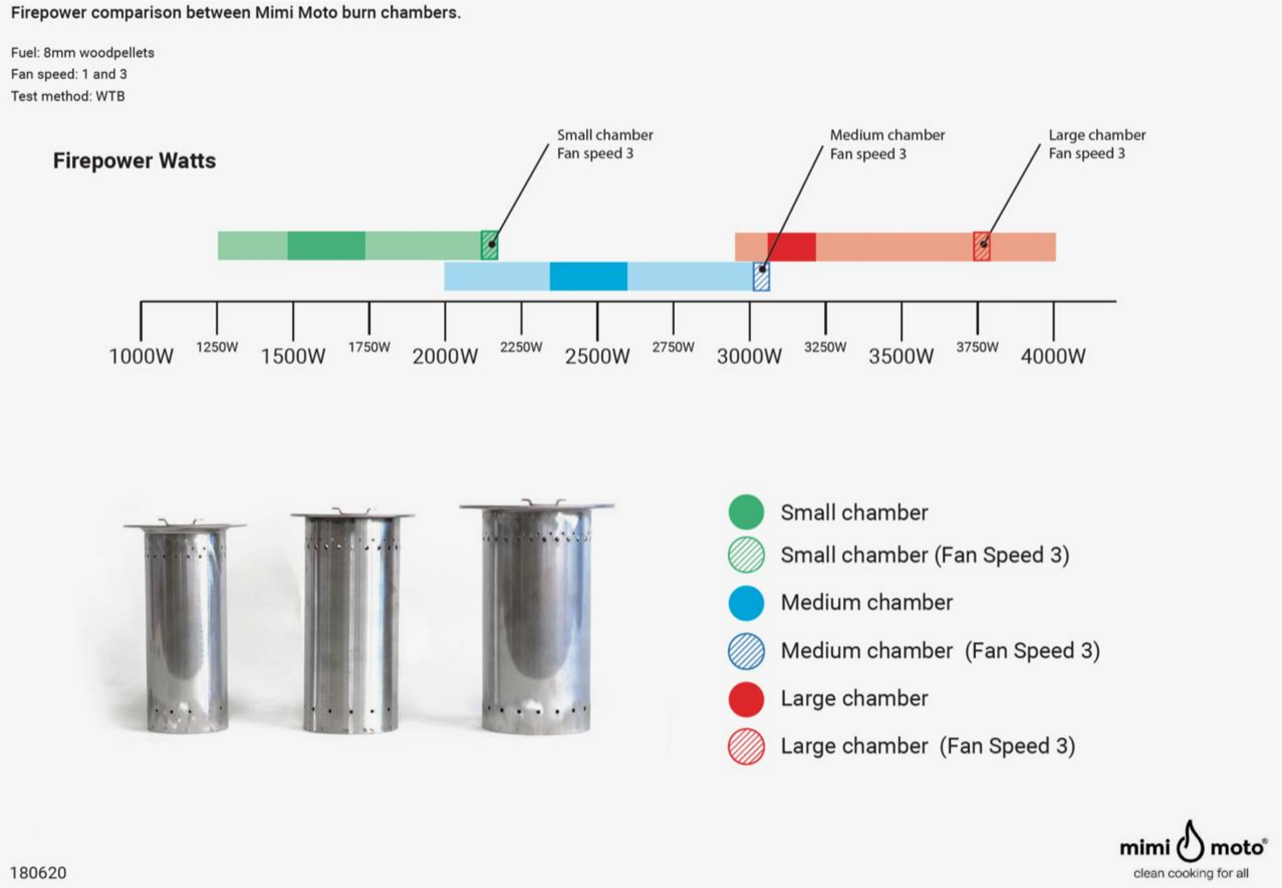
The project is neither registered nor seeking registration under another carbon scheme apart from the Gold Standard.

A.3 Technologies and/or measures

EcoSafi offers an "energy utility" business model as a replacement for cooking (primarily) with charcoal. In this model, the clean cookstoves are the infrastructure - they require wood pellet fuel for functionality, and fuel pellets are the energy; and one is useless without the other. By creating a system that incorporates both the fuel and the stove, EcoSafi creates scalable demand for an innovative product while maximizing the health, environmental, social and other sustainable development impacts of clean cooking.

Under this model, the stoves are distributed to customers who are likely to benefit the most from the intervention. The stoves are leased with a small, refundable deposit, so long as the users continue to purchase minimum quantities of pellet fuel to use in them. In return, users receive a guarantee of lifetime repair or replacement of the equipment. This reduces the adoption risk, while allowing even low-income households to benefit.

The project will primarily utilize the Mimi Moto stove, designed in the Netherlands and manufactured in China. It features a simple, intuitive design, and offers robust performance and a strong warranty. It has a certified thermal efficiency of 54.7%.



It is possible that the project will introduce a different type of stove during the crediting period. However, each stove used in the project will also be a gasifier-type clean cookstove that uses biomass fuel pellets and that is rated ISO Tier 4+ on all metrics, with an efficiency at or above 50.00% and an expected lifetime of five years or greater. In case non-Mimi Moto stoves are used by the project, relevant technical details of these stoves will be included in the relevant monitoring report(s).

Technical life of project stoves

The expected technical life of Mimimoto stoves is 5 years, based on experiences with early users in Zambia and Rwanda.

Ecosafi offers a cooking service model to all users included in this project which includes free repair/replacement services (as specified in user contracts). Repairs/replacements are free during all the usage period with no limit. Therefore, the clean cookstove's lifetime can also be clearly longer than 5 years.

Due to constant fuel supply, Ecosafi is constantly in touch with users and will thus notice if technical problems occur. However, if technical issues occur in between fuel sales, customers can easily contact the company and its employees via phone or in-person at an EcoSafi branch.

Thermal efficiency over time

The Advanced Biomass Combustion Lab (ABL) of Colorado State University validated that no decrease of efficiency is to be expected[6]. (*"Based on the testing of the Mimi Moto stove, along with an understanding of the construction and operation of the stove, we concur with the supplier's statement regarding its expected non-decrease in efficiency over time."*)

This is because failures would stop the pyrolysis process and would therefore entirely stop stove operation. In such a case, no pellets would be consumed and no ER claimed until a new stove is provided by EcoSafi and pellet use resumes as usual.

Assumptions made by ABL include:

- *"Air holes in burn chamber are cleaned regularly by the user."*
- Extensive field experience in Kenya, Rwanda and Zambia shows that there is no need to clean the airholes, because pellets used have an ash melting point which is higher than the temperatures achieved in the stove. For this reason, there is never any "sintering", i.e. formation of slag-like and hard deposits.
- *The stove chambers are replaced at the end of useful life*
- Replacement of burning chambers is part of the maintenance program.

Positive contributions to SDGs:

SDG 13, Climate Action: Reduction of GHG emissions by replacing non-renewable cooing fuels (mainly charcoal) with highly efficient wood pellets.

[6] Certificate provided to the VVB

SDG 3, Good health and wellbeing: Avoidance of negative health impacts due to smoke from conventional charcoal stoves – the project stoves burn with a clean flame comparable to LPG.
SDG 5, Gender Equality: The project stoves are much faster to light than conventional charcoal stoves, saving time particularly to women who mostly do the cooking.
SDG 7: Affordable and clean energy: Clean gasifier cookstoves are made available to customers; and due to the high efficiency, they can reduce cooking fuel expenses.
SDG 8: Decent work and economic growth: The project generates employment in stove distribution and in the pellet supply chain, among others.

A.4 Scale of the project

The project is microscale; the stoves that will be used in this project are expected to represent an annual CO2 reduction of 10,000 tCO2e or less.

A.5 Funding sources of project

The project is funded privately by EcoSAfi through the Better Cooking company. An important part of revenues that are needed to make the project viable will be covered by sales of carbon credits.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

- Gold Standard "Methodology for metered & measured energy cooking devices"[7], version 1.2.
- CDM tool "Calculation of the fraction of non-renewable biomass" (V.4.0)[8]
- Standardized baseline Grid Emission Factor for the Republic of Kenya Version 01.0[9] ASB0050-2020_PSB0055.pdf (unfccc.int)

B.2. Applicability of methodology (ies)

Criterion under 2.2.1 of MECD	Justification
a) Project shall choose a technology design that has predictable performance in that it is proven to be efficient and durable under field conditions; for fuel-based cookstoves, the rated thermal efficiency shall be higher than the baseline technology and at least 40%.	The certified thermal efficiency of the Mimi Moto stove is 54.7%. The main are charcoal and LPG stoves with efficiencies of 20% and 51%.
b) The technology shall have continuous useful energy output of less than 150kW per unit, where "continuous useful energy output" is defined (MECD 1.1.1 a.) as the energy transferred to the contents of a cooking vessel, including the sensible heat that raises the temperature of the contents of the cooking vessel and the latent heat of evaporation of water from the cooking	The maximum useful energy output of the Mimi Moto stove is 4.5 kW (see graphic in A.3 provided by the stove producer, mentioning a maximum of 4kW, to which 0.5kW are added to consider possible exceptions).

7 Methodology for metered & measured energy cooking devices – Gold Standard for the Global Goals
8 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf
9 https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20201230125121808/ASB0050-2020_PSB0055.pdf

vessel, divided by the time of the operation of the cooking task.	
c) The project activity is implemented by a project developer and can include additional project participants. However, the individual households and institutions may be represented collectively by community organisations, etc., but do not individually act as project participants.	It is confirmed that individual households and other stove users will not act as project participants.
d) The project developer must design incentive mechanism(s), which should be effective as fast as possible, for the elimination of any inefficient baseline stoves that are replaced entirely by the project cooking device(s) and describe the incentive mechanism(s) in the PDD/VPA-DD at the time of validation.	The most effective incentive is the cost advantage of cooking with pellets on the Mimi Moto stove in comparison to charcoal stoves. In addition, the resulting clean and comfortable cooking environment represents an additional strong incentive.
e) To avoid double counting or double claiming, the project developer must: i. clearly communicate its ownership rights and intention of claiming the emission reductions resulting from the project activity to the following parties by contract or clear written assertions in the transaction paperwork: all other project participants; project technology manufacturers; and retailers of the project technology or the renewable fuel in use; and ii. inform and notify the end users that they cannot claim emission reductions from the project, and iii. exclude from the project activity, project devices included in any other voluntary market or CDM project activity/PoA, and strive not to displace the cooking technologies of another CDM or	i. The transfer of ownership has been discussed during the LSC and it is confirmed by each individual user in the user agreement. Pellet producers will provide statements on transfer of ownership. Stoves are distributed by Ecosafi, without intermediaries. ii. The same is also confirmed in the user agreement iii. This is ensured by the use of a three-tiered system for tracking customers, stoves and pellet sales (including assignment of a unique identification number for each customer receiving a device under this project, which is linked to the serial number for each clean cookstove they use as well as to the

voluntary project/PoA. See section 3.11, avoidance of double counting or double claiming with other mitigation actions, for details on this demonstration.	mobile money transaction numbes associated with their pellet purchases). (SeeA.1.1.C. and B.7.3).
f) Under this methodology, emission reductions cannot be claimed for fuel-switch only, so proposed project activities also need to introduce alternative technologies, i.e. technology switch is also involved.	The project introduces micro gasifier stoves, a new technology.
g) For project cooking devices that use fossil fuel, only emission reductions from efficiency improvement are eligible.	Project devices use renewable/residual biomass.
h) For project cooking device that use grid electricity, only emission reductions from fuel switch and efficiency improvement are eligible.	No grid electricity is used for cooking.
i) The measured fuel or energy is used to calculate both baseline and project emissions. The project developer must have monitoring systems in place to monitor the fuel or energy consumption by all the project devices under the project to be recorded in a database, which is maintained by the project developer.	Fuel consumption of all project devices will be monitored continuously and thoroughly through fuel sales to users (see B.7.1).

Additional note on the eligibility according to the scope of the of the technology as "metered energy cooking device"

Demonstration that the project is within the Scope of MECD (2.1.1 and 2.1.2 of MECD):

2.1.1 | This methodology is applicable to project activities that introduce technologies that reduce or avoid greenhouse gas (GHG) emissions and quantify emission reductions from cooking devices through direct measurement of energy or fuel consumed, in households, communities, and/or institutions such as schools, prisons or hospitals (hereinafter referred as end-users).

Advanced pellet cookstoves are introduced that quantify emissions reductions from pellet fuel consumption (see detailed explanation below).

2.1.2 | This methodology may be applied by project developers promoting the installation of improved cooking devices, where the actual amount of energy or fuel used in the project scenario is measured directly in real-time for every device or otherwise monitored via measurement. The methodology includes the following metered cooking devices, but is not restricted to:

e. bio-ethanol cookstoves.

In the case of cooking devices using fuel, the amount of fuel purchased for cooking by each customer shall be recorded with arrangements to ensure the fuel is used for cooking and to prevent the alternative use of the fuel.

See detailed comment below. Pellet stoves are not mentioned explicitly, but they are similar to ethanol stoves since pellet fuel purchased by each customer for cooking is recorded and alternative uses are prevented (by the high prices of pellets and missing suitability for alternative uses, see B.7.3).

Detailed comment on eligibility

The definition for metered cooking devices in 1.1.1c requires devices to "record fuel or energy use directly, or through a supplementary meter with the ability to record amount of energy or fuel used for cooking over a period of time.

“Supplementary meter” is not further defined in this section, but more detailed provisions are given in the parameter table MECD 14 (*$P_{p,d,y}$, amount of fuel used in the project in by device d in year y)*,. According to these provisions, such "supplementary meters” include weighing scales applied when packaging the fuel together with sales receipts for quantities of fuel sold to stove users. Additionally, 1.1.1c, mentions as an example for eligible stoves *"… ethanol cookers when metered and sold for use in a dedicated device*" which apparently refers to metered ethanol fuel, in analogy to pellets. MECD is chosen because, in line with 2.1.2, "*the actual amount of energy or fuel used in the project scenario is measured directly in real-time for every device or otherwise monitored via measurement".* Measurement occurs according to the provision in the parameter box MECD 14 on the parameter *$P_{p,d,y}$ (amount of fuel used in the project in by device d in year y)* which requires under source of data: *"Direct measurement by metering or at sales, either on a device basis or cluster-of-devices basis."* Moreover, under *"any comment"* in the same box, it says*: "In case direct metering is not applied,* *then the fuel purchases, which are summarized on a monthly basis, are automatically captured on a continuous basis. Measurement may occur cluster-wise where project-specific retailers can clearly be assigned to customers and alternative uses are obviously excluded (if, for example, a new fuel is introduced specifically for project devices)." Moreover, as mentioned under B.7.1, for parameter $P_{p,d,y}$, the project plans to include a small number stoves with remote measurement of stove running time, delivering additional real-time data that will be used to cross-check that fuel consumption rates from sales are correct.*

Additional note on technical life of project technology

Section 2.3.2 of MECD requires a description of measures to ensure the durability of the technology over the time credits can be claimed. Specifications on stove life-time and repair/replacement are provided in section A.3 of this document.

B.3. Project boundary

	Source	GHGs	Included?	Justification/Explanation
Baseline scenario	Delivery of thermal energy from charcoal, LPG, kerosene and potentially use of electricity for cooking	CO_2	Yes	Important source of emissions
		CH_4	Yes	Important source of emission
		N_2O	Yes	Minor source of emissions
		…		
	Production of fuel, transport of fuel	CO_2	Yes	Important source of emissions
		CH_4	Yes	Important source of emissions
		N_2O	Yes	Minor source of emissions
		…		
Project scenario	Production of pellets from residual biomass and transport (electricity and fuel consumption)	CO_2	Yes	Minor source of emissions
		CH_4	Yes	Minor source of emissions
		N_2O	Yes	Minor source of emission
		…		
	Delivery of thermal energy from pellets in clean cookstoves	CO_2	Yes	Minor source of emissions
		CH_4	Yes	Minor source of emissions
		N_2O	Yes	Minor source of emissions

B.4. Establishment and description of baseline scenario

In the absence of the project, households and institutions (e.g. small restaurants) cook on traditional stoves with conventional fuels, mainly charcoal and LPG. It is very common to stack these fuels. The project targets users with charcoal as the primary fuel. Preliminary evaluations have shown that the Mimi Moto stove is generally used to replace charcoal, due to the following reasons:

- Charcoal is typically used for cooking the main meals, while LPG is used for minor tasks such as quick preparation of hot beverages.
- Charcoal is less convenient for cooking in comparison to LPG and therefore households and institutions have a higher tendency to replace it.
- Pellets are available in small packages similar to those used for charcoal, while LPG is only available in larger cylinders at a higher price.

Different fuel shares are being used in the baseline, and pattens vary a lot in Nairobi. It would be impossible to standardize fuel consumption patterns in a generalized baseline. The most accurate solution are individual baselines.

Therefore, when subscribing to the project, Mimo Moto stove users indicate their baseline fuel usage pattern, and emission reductions are calculated for each stove user's personal baseline.

B.5. Demonstration of additionality

Use this table for Automatic Additionality Only – delete if N/A

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	GS4GG Community-Services Activity Requirements section 4.1.9 (c).
Describe how the proposed project meets the criteria for deemed additionality.	The proposed project is a Microscale activity as annual generation of emission reductions is limited to a maximum of 10,000 tonnes of CO_2eq

B.5.1 Prior Consideration

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B.5.2 Ongoing Financial Need

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B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

SUSTAINABLE DEVELOPMENT GOALS TARGETED	MOST RELEVANT SDG TARGET	SDG IMPACT **INDICATOR (PROPOSED OR SDG INDICATOR)**
13 Climate Action (mandatory)	N/A	Reduced GHG emissions
3 Good health and wellbeing	3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	• Avoided kitchen smoke exposure • Reduced reported health impacts of kitchen smoke
5 Gender equality	5.4 Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate	Time saved by women from faster cooking
7 Affordable and clean energy	7.1 By 2030, ensure universal access to affordable, reliable and modern energy services	Access to clean cooking technology
8 Decent work and economic growth	8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	Employment generation

B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact

Baseline Emissions

According to the methodology MECD, a baseline emission factor per unit of useful cooking energy will be established, based on the fuel mix in the pre-project case and baseline stove efficiencies. Baseline emissions will then be calculated by multiplying the useful cooking energy provided by project devices (based on monitoring of pellet consumption) with the baseline emission factor.

Baseline Emission Factor:

$$EF_{b,useful} = \sum_k \left(\sum_{i,j} P_{b,i,j} \times Percentage\ of\ fuel\_i \times (EF_{b,i,CO2} \times fNRB_{i,y} + EF_{b,i,non-CO2}) \times NCV_{b,i} \right)_k \div \sum_k \left(\sum_{i,j} P_{b,i,j} \times Percentage\ of\ fuel\_i \times NCV_{b,i} \times \eta_{b,i,j} \right)_k$$

Eq 1 of MECD

Where:

$EF_{b,useful}$	=	Baseline emissions factor (tCO_2e per TJ of useful energy), for entire mix of baseline cooking energy
$P_{b,i,j}$	=	Amount of baseline fuel *i* used in device *j* in year *y* (tonnes)
$EF_{b,i,CO2}$	=	Emission factor of the baseline fuel *i* (tCO_2e/tonne)
$EF_{b,i,non-CO2}$	=	Emission factor of the baseline fuel *i* (tCO_2e/tonne)
$fNRB_{i,y}$	=	Non-renewability status of woody biomass fuel i during year y
$NCV_{b,i}$	=	The net calorific value of the baseline fuel type *i* (TJ/tonne)
$\eta_{b,i,j}$	=	Efficiency of baseline device *j* with fuel i (fraction)
$Perentage\ of\ fuel\_i_i$	=	Percentage of fuel type I in baseline situation (%)
k	=	Household *k* from the target population

j	=	Baseline devices *j*
i	=	Baseline fuel *i*

Parameter values for Eq. 1 are obtained as described in the parameter tables (B.6.2). On $P_{b,d,j}$, explanations are given here below. On ***fNRB***, explanations have been moved to the bottom of this section B.6.1., for better readability.

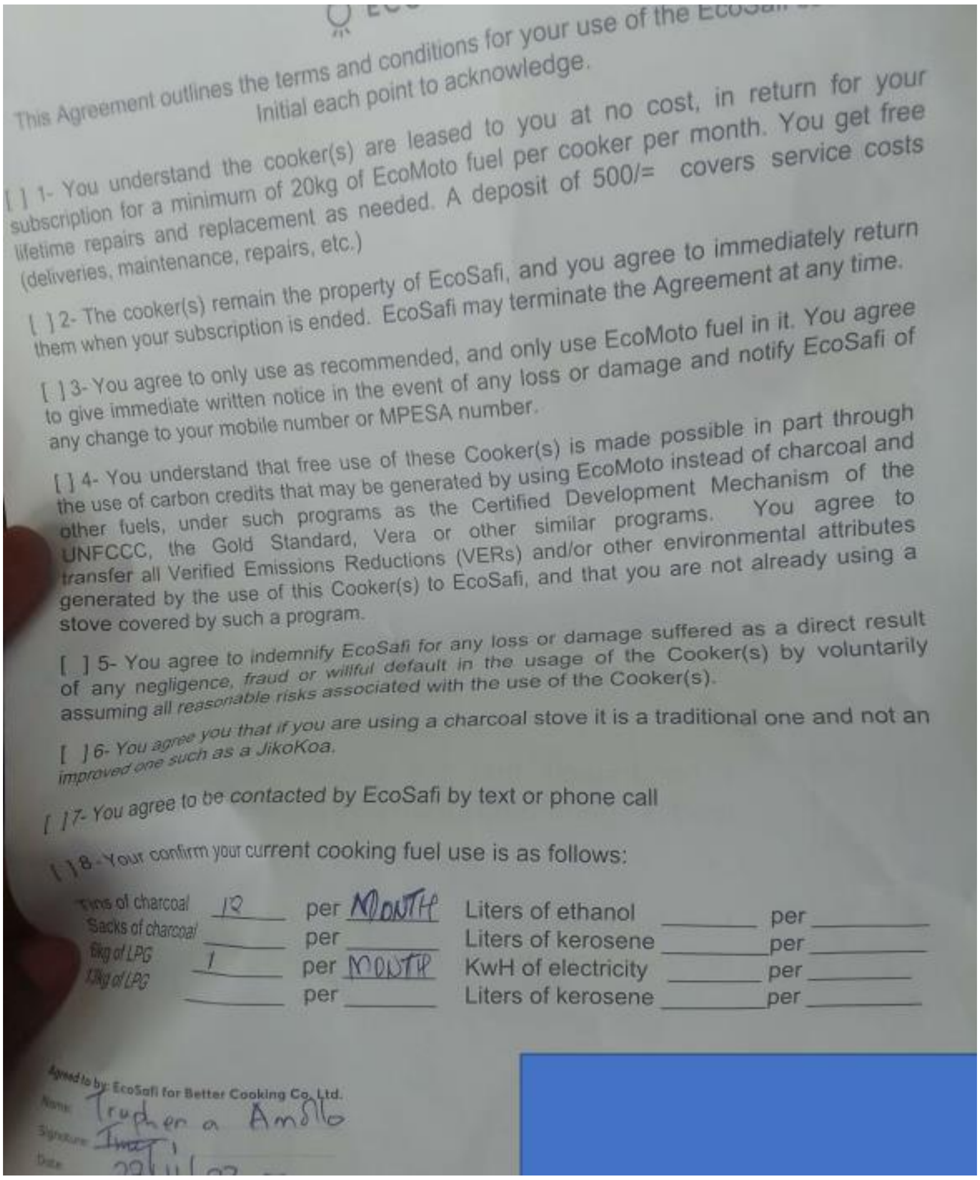
$P_{b,d,j}$

Baseline fuels used by the target group in urban Nairobi include charcoal, LPG and kerosene. Electricity is used very occasionally; therefore, users of electric cookstoves are excluded from emission reduction calculation. Also ethanol users who are already registered under another carbon program are excluded. Firewood is not used by the target group; however it is considered in calculations in case the project expands to peri-urban aeras where firewood usage is more common.

Baseline fuel mixes vary a lot in Nairobi and therefore cannot be captured in a generalized baseline. Therefore, an individual baseline is established for each user. When subscribing to the project, each user indicates her/his baseline fuel usage pattern. Experience has shown that results are most accurate if stove users are asked for the quantity of typical fuel packaging sizes they use in a defined period of time. Typical packaging sizes include:

- Tins or bags of charcoal (containing 1.4 and 50kg of charcoal).
- LPG cylinders of 6 and 13 kg, rarely 3kg cylinders also.
- Kerosene is usually quantified by the amount of money paid for it. The price in late 2022 is 150 KSH/l

  Below two examples of signed user contracts with personal baseline fuel consumption:



This Agreement outlines the terms and conditions for your use of the EcoSafi ...
Initial each point to acknowledge.

[ ] 1- You understand the cooker(s) are leased to you at no cost, in return for your subscription for a minimum of 20kg of EcoMoto fuel per cooker per month. You get free lifetime repairs and replacement as needed. A deposit of 500/= covers service costs (deliveries, maintenance, repairs, etc.)

[ ] 2- The cooker(s) remain the property of EcoSafi, and you agree to immediately return them when your subscription is ended. EcoSafi may terminate the Agreement at any time.

[ ] 3- You agree to only use as recommended, and only use EcoMoto fuel in it. You agree to give immediate written notice in the event of any loss or damage and notify EcoSafi of any change to your mobile number or MPESA number.

[ ] 4- You understand that free use of these Cooker(s) is made possible in part through the use of carbon credits that may be generated by using EcoMoto instead of charcoal and other fuels, under such programs as the Certified Development Mechanism of the UNFCCC, the Gold Standard, Vera or other similar programs. You agree to transfer all Verified Emissions Reductions (VERs) and/or other environmental attributes generated by the use of this Cooker(s) to EcoSafi, and that you are not already using a stove covered by such a program.

[ ] 5- You agree to indemnify EcoSafi for any loss or damage suffered as a direct result of any negligence, fraud or willful default in the usage of the Cooker(s) by voluntarily assuming all reasonable risks associated with the use of the Cooker(s).

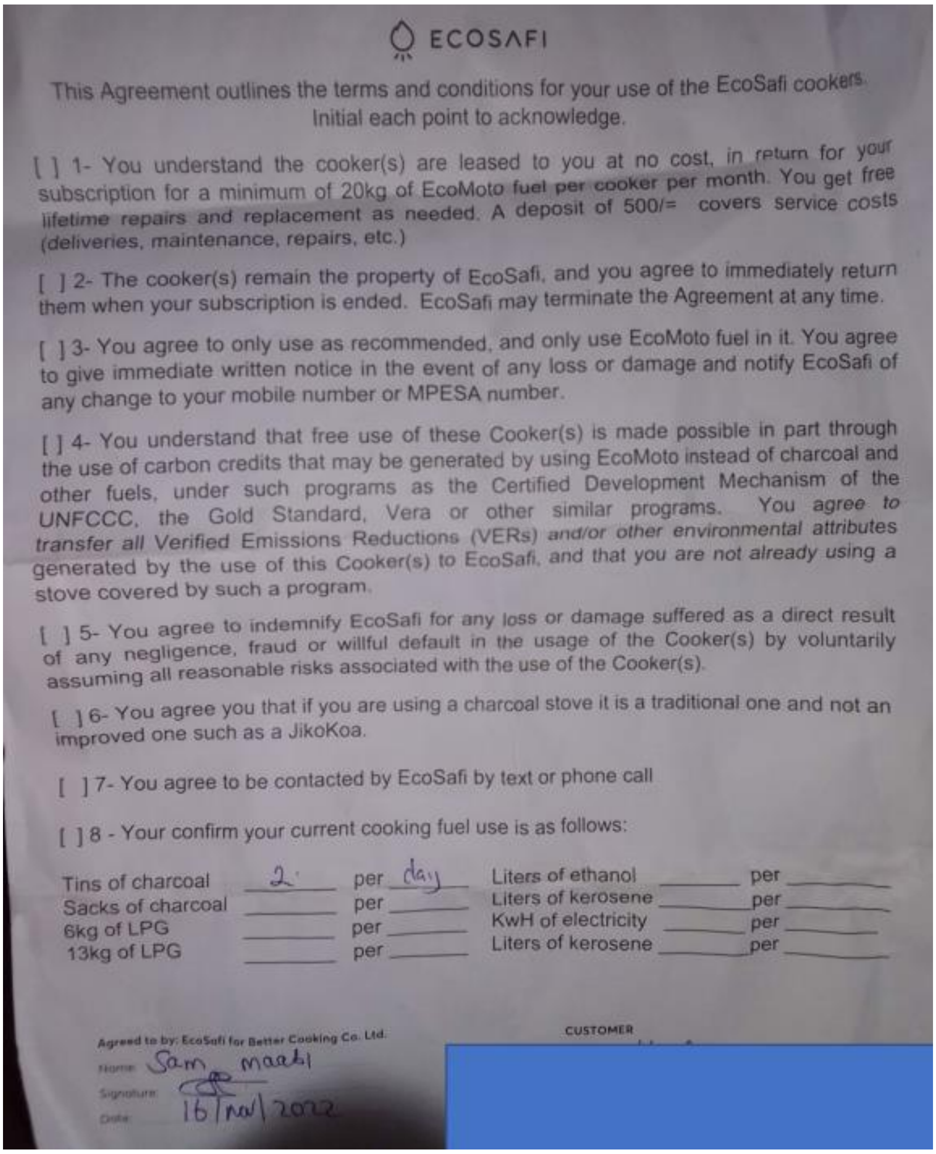
[ ] 6- You agree you that if you are using a charcoal stove it is a traditional one and not an improved one such as a JikoKoa.

[ ] 7- You agree to be contacted by EcoSafi by text or phone call

[ ] 8- Your confirm your current cooking fuel use is as follows:

Tins of charcoal	12	per MONTH	Liters of ethanol		per	
Sacks of charcoal		per	Liters of kerosene		per	
6kg of LPG	1	per MONTH	KwH of electricity		per	
13kg of LPG		per	Liters of kerosene		per	

Agreed to by: EcoSafi for Better Cooking Co. Ltd.
Name: Truphena Amollo
Signature:
Date:



ECOSAFI

This Agreement outlines the terms and conditions for your use of the EcoSafi cookers.
Initial each point to acknowledge.

[ ] 1- You understand the cooker(s) are leased to you at no cost, in return for your subscription for a minimum of 20kg of EcoMoto fuel per cooker per month. You get free lifetime repairs and replacement as needed. A deposit of 500/= covers service costs (deliveries, maintenance, repairs, etc.)

[ ] 2- The cooker(s) remain the property of EcoSafi, and you agree to immediately return them when your subscription is ended. EcoSafi may terminate the Agreement at any time.

[ ] 3- You agree to only use as recommended, and only use EcoMoto fuel in it. You agree to give immediate written notice in the event of any loss or damage and notify EcoSafi of any change to your mobile number or MPESA number.

[ ] 4- You understand that free use of these Cooker(s) is made possible in part through the use of carbon credits that may be generated by using EcoMoto instead of charcoal and other fuels, under such programs as the Certified Development Mechanism of the UNFCCC, the Gold Standard, Vera or other similar programs. *You agree to transfer all Verified Emissions Reductions (VERs) and/or other environmental attributes generated by the use of this Cooker(s) to EcoSafi, and that you are not already using a stove covered by such a program.*

[ ] 5- You agree to indemnify EcoSafi for any loss or damage suffered as a direct result of any negligence, fraud or willful default in the usage of the Cooker(s) by voluntarily assuming all reasonable risks associated with the use of the Cooker(s).

[ ] 6- You agree you that if you are using a charcoal stove it is a traditional one and not an improved one such as a JikoKoa.

[ ] 7- You agree to be contacted by EcoSafi by text or phone call

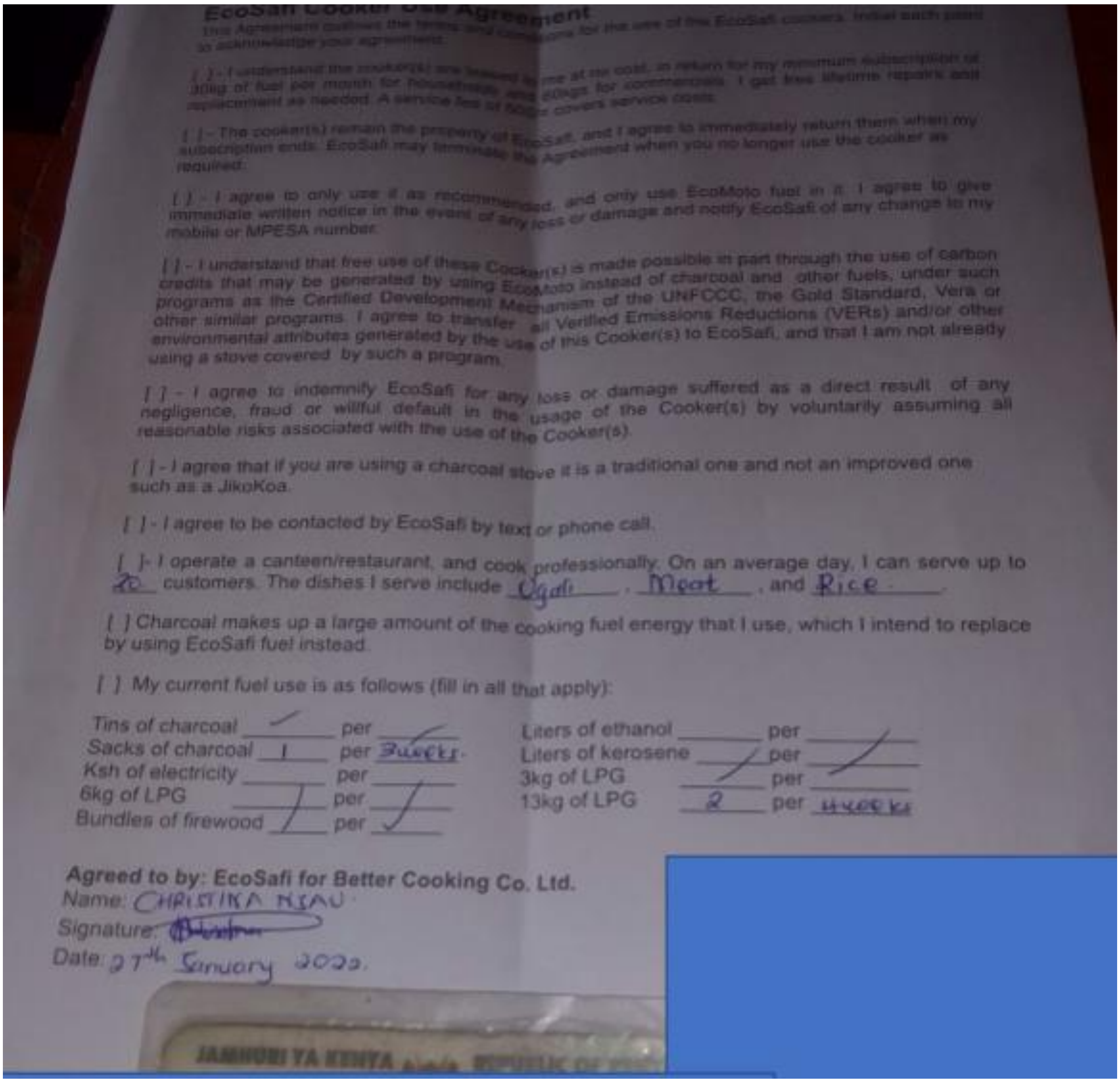
[ ] 8 - Your confirm your current cooking fuel use is as follows:

Tins of charcoal	2	per	day	Liters of ethanol		per	
Sacks of charcoal		per		Liters of kerosene		per	
6kg of LPG		per		KwH of electricity		per	
13kg of LPG		per		Liters of kerosene		per	

Agreed to by: EcoSafi for Better Cooking Co. Ltd.
Name: Sam maabi
Signature:
Date: 16/nov/2022

CUSTOMER
[redacted]

Besides households, also small restaurants are targeted. Such institutional users additionally indicate the number of meals served and dishes offered:



EcoSafi Cooker Use Agreement

This Agreement outlines the terms and conditions for the use of the EcoSafi cookers. Initial each point to acknowledge your agreement.

[ ] - I understand the cooker(s) are leased to me at no cost, in return for my minimum subscription of 30kg of fuel per month for households and 60kgs for commercials. I get free lifetime repairs and replacement as needed. A service fee of 50Ksh covers service costs.

[ ] - The cooker(s) remain the property of EcoSafi, and I agree to immediately return them when my subscription ends. EcoSafi may terminate the Agreement when you no longer use the cooker as required.

[ ] - I agree to only use it as recommended, and only use EcoMoto fuel in it. I agree to give immediate written notice in the event of any loss or damage and notify EcoSafi of any change to my mobile or MPESA number.

[ ] - I understand that free use of these Cooker(s) is made possible in part through the use of carbon credits that may be generated by using EcoMoto instead of charcoal and other fuels, under such programs as the Certified Development Mechanism of the UNFCCC, the Gold Standard, Vera or other similar programs. I agree to transfer all Verified Emissions Reductions (VERs) and/or other environmental attributes generated by the use of this Cooker(s) to EcoSafi, and that I am not already using a stove covered by such a program.

[ ] - I agree to indemnify EcoSafi for any loss or damage suffered as a direct result of any negligence, fraud or willful default in the usage of the Cooker(s) by voluntarily assuming all reasonable risks associated with the use of the Cooker(s).

[ ] - I agree that if you are using a charcoal stove it is a traditional one and not an improved one such as a JikoKoa.

[ ] - I agree to be contacted by EcoSafi by text or phone call.

[ ] - I operate a canteen/restaurant, and cook professionally. On an average day, I can serve up to 20 customers. The dishes I serve include Ugali, Meat, and Rice.

[ ] Charcoal makes up a large amount of the cooking fuel energy that I use, which I intend to replace by using EcoSafi fuel instead.

[ ] My current fuel use is as follows (fill in all that apply):

Tins of charcoal ✓	per	Liters of ethanol	per
Sacks of charcoal 1	per 3weeks	Liters of kerosene	per
Ksh of electricity	per	3kg of LPG	per
6kg of LPG	per	13kg of LPG 2	per 4weeks
Bundles of firewood	per		

Agreed to by: EcoSafi for Better Cooking Co. Ltd.

Name: CHRISTIKA NJAU

Signature:

Date: 27th January 2022.

JAMHURI YA KENYA REPUBLIC OF KENYA

Data from the first >1,000 users show that in average, households consume around 250kg charcoal and 100kg LPG per year, while restaurants consume in average 2t of charcoal and 250kg of LPG. Kerosene is only used marginally.

A comparison with The Kenya Cooking Sector Study[10] shows that individual baseline surveys lead to conservative results. This Cooking Sector Study mentions fuel

---

[10] REPUBLIC OF KENYA, MINISTRY OF ENERGY (2019): Kenya Cooking Sector Study: Assessment of the Supply and Demand of Cooking Solutions at the Household level. Abridged Report https://www.fao.org/forestry/energy/catalogue/search/detail/en/c/1306942/

consumption by user type (figure 5 on page 17)[11]. Even for primary LPG users, a charcoal consumption of approx. 330kg/a is reported here, with LPG consumptions of 84 kg/a (still regarded primary LPG consumers due to higher NCV and stove efficiency for LPG). Even 630kg/a charcoal and 56 kg/a LPG is reported for primary charcoal users. It can thus be seen that the accurate results from the individual survey lead to lower baseline emissions and therefore are conservative.
This difference becomes even more clear if comparing with the CDM standardized baseline for Kenya[12] that indicates a woody biomass[13] consumption of 830kg per year and person in urban areas.

The table below shows EcoSafi's survey-based baseline charcoal usage figures compared to other relevant sources (consumption per household and year):

Fuel/Source:	CDM Standardized Baseline for Kenya	Kenya Cooking Sector Study (2019): Primary Charcoal Users	Kenya Cooking Sector Study (2019): Primary LPG Users	EcoSafi (Individual Survey Averages)
Charcoal	830 kg	630 kg	330 kg	250 kg

An ethanol cooking project that targets the same user group in urban Nairobi and was recently issued carbon credits by the Gold Standard (GS11440) even assumes a 6.82t/a (1,71t/a per person) biomass equivalent for baseline fuels, without even mentioning LPG.
Due to high prices of fossil fuel, there is also a recent tendency of increasing charcoal usage[14].

[11] All values mentioned are estimated from the diagram by multiplying the values indicated per capita and month by 3.9 (household size) and 12 (months in a year).
[12] Standardized baseline on woody biomass consumption for household cookstoves in Kenya (valid till 2020) https://cdm.unfccc.int/methodologies/standard_base/2015/sb103.html
[13] Woody biomass equivalents, representing a combination of charcoal and firewood.
[14] https://www.africanews.com/2022/07/10/outrun-by-gas-prices-kenyan-families-turn-to-charcoal-to-cook/

Overall baseline emissions are calculated as per Eq. 2:

$$BE_y = EG_{p,useful,y} \times EF_{b,useful}$$ *Eq. 3 of MECD*

Where:

BE_y	=	Baseline emissions (tCO_2e) in the year y
$EG_{p,useful,y}$	=	The amount of useful energy applied in the project in year y (TJ)
EF_b	=	Baseline emissions factor (tCO_2e per TJ of useful energy)

The amount of useful energy applied in the project is derived by Eq. 4. (Eq. 3 of MECD is not applicable since no electric stoves are deployed):

$$EG_{p,useful,y} = \sum_d P_{p,d,y} \times NCV_{p,i} \times \eta_{p,d,y}$$ *Eq. 7 of MECD*

Where:

$EG_{p,d,y}$	=	The amount of fuel used in the project in by device *d* in year *y*, considering cap (mass or volume unit)
$NCV_{p,i}$	=	The net calorific value of the fuel *i* used in the project scenario in year *y*
$\eta_{p,d,y}$	=	Energy efficiency of the project device, *d* in year *y* (fraction)
d	=	Project device *d*

All parameters used to calculate Eq. 4 are monitored, as explained in B.7.1 (Monitoring Parameter tables).

Project Emissions

Since the project uses biomass-based fuel, project emissions associated with production and transport of fuel must be evaluated (3.6.2 of MECD).

Pellet production uses electricity. A conservative value of 200kWh/ton of pellets is used, according to a fact sheet of the World Bioenergy Association that gives a range from 100-200kWh/t[15]. The pellet provider confirms a value of 127kWh/t[16]. A grid emission factor of 0.5kg CO_2/kWh is conservatively chosen, clearly above the value for the

[15] www.worldbioenergy.org/uploads/Factsheet%20-%20Pellets.pdf
[16] Provided to VVB

combined margin in the latest CDM standardized baseline[17]. It is increased by 20% according to the CDM default value for TDL (average technical transmission and distribution losses, see CDM tool 5 version 3 7.2[18]).

There are no significant emissions caused by the battery-driven fan of the Mimi Moto stove[19].

Project emissions from electricity consumption are thus calculated as:

$$PE_{electricity,y} = 0.2\ \text{MWh} * P_{total,y} * 0.5\text{tCO}_2/\text{MWh}\ (1+20\%) = 0.12\text{tCO2} * P_{total,y}$$

Where:

$PE_{electricity,y}$ = Project emissions caused by electricity consumption for pellet production

$P_{total,y}$ = Total amount of pellets consumed in project devices in a monitoring period

Transport emission have to be considered according to 3.6.3 c) of MECD. The pellet factory of the usual supplier is situated in Migori county, 370 km from Nairobi. Transport emissions are considered in a generalized way by applying an emission factor given in the CDM tool "Project and leakage emissions from transportation of freight, version 1.1" [20]. In section 5.3, default values of 129g CO2/tkm are given for heavy vehicles[21]. If

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[17] https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20201230125121808/ASB0050-2020_PSB0055.pdf
[18] am-tool-05-v3.0.pdf (unfccc.int)
[19] The battery is often charged with solar power; therefore, no grid electricity is required. Moreover, the battery can be used to charge a phone, for example, implying replacement of grid energy. Even if grid electricity is used, the consumption is negligible, leading to a consumption of approximately 1kWh per year (corresponding to 0.5kg of CO2).
[20] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-12-v1.1.0.pdf
[21] A statement of the pellet producer is provided to the VVB, clarifying that transport is hired with trucks ("Pellet producer VUMA confirmation production and price").

multiplying with 370km, project emissions of 47.7kg CO2/t is obtained. This is conservatively set to 0.05tCO_2 per ton of biomass.

Since transport is contracted[22], it is plausible that vehicles will return with loads from non-EcoSafi customers, therefore, one-way trips can be assumed.

$$PE_{transport,y} = 0.05tCO_2 * P_{total,y}$$

Where:

$PE_{transport,y}$ = Project emissions caused by transport of pellets

$P_{total,y}$ = Total amount of pellets consumed in project devices in a monitoring period

Furthermore, pellets cause CO_2 emissions when burned in the project stove. Pellets of the usual supplier are made from bagasse which is waste biomass. Where raw biomass other than bagasse or sawdust is used that cannot be deemed waste biomass, project emissions are calculated by applying the same emission factors that applies for firewood consumption in the baseline.

$$PE_{combustion,pellets,y} = P_{p,pellets,total,} * conversion_{wood\_pellets} * share_{non\text{-}waste} *$$
$$(EF_{b,wood,\ CO2} * f_{NRB} + EF_{b,wood,non\text{-}CO2})$$

Where:

$PE_{combustion,\ pellets,y}$ = Emissions caused by fuel burned in the project stove

$conversion_{wood\_pellets}$ = Wood-to-pellet conversion factor (1.2, meaning that 12kg dry woody biomass are required to produce 1t of dry pellets)

$share_{non\text{-}waste}$ = Percentage of the raw material not deemed waste biomass (equivalent to non-renewable), monitored.

$EF_{b,wood}$ = Emission factor for wood 1.89t CO2/t, based on 121.46tCO2/TJ[23]) times NCV_{wood} (0.0156TJ/t[24])

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[22] Pellet producer confirmation provided to VVB
[23] $EF_{wood,per\ energy}$ (121.46tCO2/TJ), as per table 2.5 of Chapter 2 of Vol. 2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories (112,000 kgCO2/TJ, 300kgCH4/TJ and 4kgN2O/TJ), and AR 5 GWPs (28 for CH4 and 265 for N2O)
[24] Table 1.2 of Chapter 1, Introduction to Volume 2 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories

$PE_y = PE_{electricity,y} + PE_{transport,y} + PE_{combustion,pellets,y}$

Leakage Emissions

MECD makes reference to 3.11 of RECH (Gold Standard Methodology on Reduced Emission from Cooking and Heating). Here an approach is described for a systematic assessment of leakage.

a. The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or with a higher intensity than would have occurred in the absence of the project.

Baseline stoves will often remain in use, while continued usage is automatically monitored through fuel consumption. If that should not be the case, it is not plausible that other households would use these devices and in consequence increase biomass consumption. Since biomass is the cheaper cooking option, non-availability of conventional biomass stoves is not a limiting factor.

b. Members of the population who do not participate in the project, and previously used lower emitting energy sources, instead use the non-renewable biomass or fossil fuels saved under the project activity.

The project is too small to have a significant effect on the charcoal market, given that there are > 1.4 million households in Nairobi only[25].

c. The project significantly reduces the NRB fraction within an area where other GHG mitigation project activities account for NRB fraction in their baseline scenario.

The influence of the project on fNRB is insignificant, given that there are more than 1.4 million household sin Nairobi only.

d. The project population compensates for loss of the space heating effect of inefficient technology by adopting some other form of space heating or by retaining some use of inefficient technology.

[25] htpps://housingfinanceafrica.org/documents/2019-kenya-population-and-housing-census-reports/

The climate in Nairobi is very stable, average temperatures are >20° in each month of the year. And in any case, baseline stoves will often remain in use, while continued usage is automatically monitored through fuel consumption.

e. By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution with this technology by households who commonly used a technology with relatively lower emissions.

All users are known to the project, and both stoves and fuel are sold centrally. Moreover, the baseline survey considers average usage of different technologies (of which the project technology is still the most efficient).

In conclusion, no leakage emissions have to be considered.

Calculation of Emission Reductions

$$ER_y = BE_y - PE_y - LE_y$$ *Eq. 10of MECD*

Where:

ER_y	=	Emission reductions in year y (t CO2e/yr)
BE_y	=	Baseline emissions in year y (t CO2e/yr)
PE_y	=	Project emissions in year y (t CO2/yr)
LE_y	=	Leakage emissions in year y (t CO2/yr)

Determination of the fraction of non-renewable biomass

As per the CDM tool "Calculation of the fraction of non-renewable biomass" (V.4.0)[26].

The overall equation to be applied is:

$$fNRB = \frac{NRB}{NRB + RB}$$ ***(equation 1 of tool)***

where:

fNRB: *Fraction of non-renewable biomass (fraction or %)*

NRB: *Quantity of non-renewable biomass (t/yr)*

RB: *Quantity of renewable biomass (t/yr)*

The quantity of non-renewable biomass is derived by the equation given by:

$$NRB = H - RB$$ (***equation 2 of tool)***

Where:

H: *Total annual consumption of wood in the absence of the project activity in tons/yr*

Determination of the total wood consumption *H*

$$H = HW * N + CE + NE - RB$$ (***equation 3 of tool)***

Where:

*HW * N:* *Total household's fuelwood consumption in tons. The tool par. 15 allows to use an aggregated value for consumption per household (HW) times the number of households (N).*

CE: *Commercial woody biomass consumption for energy applications (e.g., commercial, industrial or institutional uses of woody biomass in ovens, boilers*

[26] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf

etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

NE: *Commercial woody biomass consumption for non-energy applications (e.g. construction, furniture) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)*

Determination of the quantity of renewable biomass (*RB*)

(***equation 4 of tool***)

$$RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest})) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other}))$$

Where:

$MAI_{forest,i}$: Mean Annual Increment of woody biomass growth per hectare in sub-category i of forest areas (t/ha/yr)

$MAI_{oother,i}$: Mean Annual Increment of woody biomass growth per hectare in sub-category i of other wooded land areas (t/ha/yr)

$F_{forest,i}$: Extent of forest in sub-category i (ha)

$F_{other,i}$: Extent of other wooded land in sub-category i (ha)

P_{forest} : Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (ha)

P_{orther} :Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other wooded land areas (ha)

i : Sub-category i of forest areas and other wooded land areas

Detailed sources and calculations are provided in the Excel Sheet on fNRB calculation.

NRB = H - RB

= 56.0 mio t/y – 5.5 mio t/y

= 50.5 mio t/y

$$fNRB = \frac{NRB}{NRB + RB}$$

= 50.53 / 55.99

The results of the calculation is 90.25%.

According to Bailis et al. (2015)[27], Kenya is part of a hot spot in Eastern Africa (see also figure 4 of the article). The map presented by Bailis et al. (2015) shows values up to 80% for Kenya. A high value seems plausible in the light of high importance of charcoal among the project's target group. In difference to firewood, charcoal is mostly produced from standing trees; since it is easier to transport, it is often brought from relatively dry areas where hard wood is found. It has thus a higher impact on forests than firewood which is often collected from dead biomass.

However, Bailis et al (2015) also makes clear that "past efforts to quantify woodfuel sustainability failed to provide credible results" while proposing an improved method of calculating fNRB based on a spatially-explicit assessment of pan-tropical woodfuel supply and demand because "carbon offset projects…are probably overstating the climate benefits of improved stoves."

Bailis et al (2015) apply different scenarios when undertaking the calculations and estimates using the assessment method. The Berkeley Carbon Trading Project (BCTP) at the University of California, Berkeley,[28] recommends that project developers use Bailis et al. 2015's "Scenario B-Low Yield" values at the subnational level if the project's scope is not national.

Although EcoSafi operates primarily in Nairobi, the Berkeley Carbon Trading Project also recommends that if "the baseline includes charcoal that is transported from throughout the country, then a national level fNRB may be more appropriate, even if the project is only operating in a specific sub-national area." Because EcoSafi targets primary charcoal users located in the capital of Nairobi, which indeed is supplied from charcoal transported from throughout the country, the expected fNRB for Kenya of 63.9% as calculated by Bailis et al (2015) in "Scenario B-Low Yield" is used.

This is significantly lower than the figure already calculated for the project above and lower than other GS registered projects, as shown in the table below:

27 https://www.nature.com/articles/nclimate2491/

[28] The Berkeley Carbon Trading Project at the Center for Environmental Public Policy (CEPP) is a research and outreach program dedicated to studying the effectiveness of carbon trading and offset programs and ensuring that this understanding informs program design. https://gspp.berkeley.edu/research-and-impact/centers/cepp/projects/berkeley-carbon-trading-project

Factor/Project	GS11603	GS11188	GS07692	GS10987	EcoSafi (Calculated)	EcoSafi (Used)
fNRB	97%	93%	92%	92%	90.25%	63.9%

However, this figure also represents a more accurate and more conservative approach to calculating fNRB and is used. As a result:

fNRB = 63.9%

Equations to calculate other SDGs than SDG 13 are explained in B.6.3, for better readability.

B.6.2 Data and parameters fixed ex ante

SDG13

Data/parameter ID	MECD 1
Data/parameter	$P_{b,i,j}$
Unit	Tonne/ year-device
Description	Amount of baseline fuel i used in device j (tonnes) in the baseline.
Source of data	Data are obtained from an individual user survey with 100% coverage
Value(s) applied	Determined individually for each user. On average, cooking energy is derived in approximately equal parts from charcoal and LPG, but with high variations between users.
Choice of data or Measurement methods and procedures	Baseline fuel mixes vary a lot in Nairobi and cannot be captured in a generalized baseline. Therefore, an individual baseline is established for each user. When subscribing to the project, each user indicates her/his baseline fuel usage pattern and confirms it by signature. Experience has shown that results are most accurate if stove users are asked for the quantity of typical fuel packaging sizes they use in a defined period of time.
Purpose of data	Emission reduction calculation
Additional comment	n/a

Data/parameter ID	MECD 2
Data/parameter	$NCV_{b,i}$
Unit	Terrajoules (TJ)/tonne of fuel
Description	The net calorific value of the baseline fuel type i
Source of data	IPCC default data, - project-relevant measurement reports, or project specific field tests. If either project-specific or project-relevant results are used, these must be cross-checked with IPCC defaults and differences shall be justified using evidence.
Value(s) applied	Wood: 0.0156 Charcoal: 0.0295 LPG: 0.0473 Kerosene: 0.0438
Choice of data or Measurement methods and procedures	IPCC default values
Purpose of data	Emission reduction calculation
Additional comment	n/a

Data/parameter ID	MECD 3
Data/parameter	$EF_{b,i,CO2}$
Unit	tCO_2e/TJ
Description	CO_2 emission factor arising from use of fuels in baseline scenario
Source of data	Wood: Methodology default Charcoal: Methodology cap LPG and Kerosene: IPDD default (Table 2.5 of Chapter 2 of Vol. 2, 2006 IPCC Guidelines)

Value(s) applied	Firewood: 112 Charcoal: 197.15 LPG: 63.1 Kerosene: 71.9
Choice of data or Measurement methods and procedures	**Charcoal:** The cap given in MECD is applied because it still represents a conservative estimate. As shown below, realistic emission factors for Kenya would still be much higher than the cap, due to low conversion efficiency in charcoal plants. This is shown by an analysis of official UNdata data, comparing hypothetical wood-to-charcoal conservation factors that are not applied under this VPA: The amount of fuelwood transformed in charcoal plants is indicated as 41,970 *1000m3[29]. If applying the FAO default density factor of 0.725t/m3[30], this corresponds to 30,438 tons. Charcoal production is indicated as 4,106 * 1000t[31]. This results in a wood-to-charcoal conversion factor of 7.4 (30,438/4,106). However, the density value applied is by far underestimated because for charcoal production, heavy woods are preferred. **Theoretically, a wood-to-charcoal conversion clearly above 6 would thus be well justified.** The corresponding theoretical wood-to-charcoal factor from **MECD cap values is only 4.5**, calculated from: [EFcharcoal,CO2 (197.15tCO2/TJ) + EFcharcoal,non-CO2 (92.29tCO2/TJ)]* NCVcharcoal (0.0295TJ/t) / [EFwood,CO2 (112tCO2/TJ) + EFwood,non-CO2 (9.46tCO2/TJ)]* NCVwood (0.0156 TJ/t) = 4.5

29 data.un.org/Data.aspx?d=EDATA&f=cmID%3aFW
30 http://www.fao.org/docrep/010/a1106e/a1106e00.htm
31 data.un.org/Data.aspx?q=charcoal&d=EDATA&f=cmID%3aCH

	This comparison of hypothetical wood-to-charcoal conversion factors shows that applying the cap of the charcoal EF is still a conservative choice.
Purpose of data	Emission reduction calculation
Additional comment	The parameter is used to calculate baseline emissions factor.

Data/parameter ID	MECD 4
Data/parameter	$EF_{b,i,CO2}$
Unit	tCO_2e/TJ
Description	CO_2 emission factor arising from use of fuels in baseline scenario
Source of data	Wood: Methodology default Charcoal: Methodology cap LPG and Kerosene: n.a.
Value(s) applied	Firewood: 9.46 Charcoal: 92.9
Choice of data or Measurement methods and procedures	**Charcoal:** See explanation under parameter MECD 3 above.
Purpose of data	Emission reduction calculation
Additional comment	The parameter is used to calculate baseline emissions factor.

Data/parameter ID	MECD 5
Data/parameter	$\eta_{b,j}$
Unit	Fraction
Description	Energy efficiency of baseline device *j* with fuel *i*
Source of data	Firewood: Methodology default value for conventional systems using woody biomass without grate or chimney Charcoal: Methodology default value for other conventional systems using woody biomass LPG: Scientific paper cited on http://aprovecho.org/the-big-picture/lpg-cooking-and-health
Value(s) applied	Firewood: 10% Charcoal: 20% LPG: 51%[32] Kerosene: 43%[33]
Choice of data or Measurement methods and procedures	According to MECD methodology.
Purpose of data	Emission reduction calculation
Additional comment	The parameter is used to determine the baseline emission factor.

[32] http://aprovecho.org/the-big-picture/lpg-cooking-and-health
[33] Shrestha, J. N. (2001). Efficiency measurement of biogas, kerosene and LPG stoves. Center for Energy Studies: Tribhuvan University Institute of Engineering.

Data/parameter ID	MECD 6
Data/parameter	*Percentage of fuel_i*
Unit	%
Description	Percentage of fuel type I in baseline situation.
Source of data	From the baseline study, two baseline user profiles are identified, consuming different shares of multiple fuels (see parameter table for $P_{b,i,j}$). Therefore, it would be redundant to apply a percentage again to these shares. The parameter is thus set to 100%.
Value(s) applied	100%
Choice of data or Measurement methods and procedures	See explanation under source of data.
Purpose of data	Emission reduction calculation
Additional comment	n/a

Data/parameter ID	No ID assigned, not described in MECD
Data/parameter	$Conversion_{wood\text{-}pellets}$
Unit	t/t
Description	Conversion factor wood-pellet. This parameter is only needed if non-renewable biomass is used to produce pellets, i.e. if $Share_{non\text{-}waste}$ >0 (see B.7.1). It indicates how much woody biomass is used as input to generate a ton of pellets.
Source of data	Generic explanation.
Value(s) applied	1.2

Choice of data or Measurement methods and procedures	When woody biomass is transformed into pellets, a mass reduction occurs, basically due to moisture reduction. Incoming biomass typically has a moisture content between 20% and 45%, since it is stored in huge piles (e.g. bagasse, sawdust at sawmills, etc.). Pellets have an average moisture content around 8%. During moisture reduction, the carbon content of biomass does not change. For the calculation of project emissions due to combustion of pellets made from non-renewable biomass ($PE_{combustion,pellets,y}$), default values for NCV_{wood} and EF_{wood} for air dried wood with up to 20% moisture are used[34]. If solely considering differences in moisture, 1kg of pellets corresponds to 1.11 kg wood (120%/108%). Normally, all biomass is transformed into pellets. However, there may be some losses of biomass in the process, due to the presence of dust and unusable bark (which is waste anyway). We conservatively assume a loss of 8.9% biomass. The resulting wood-to-pellet factor is thus set as 1.2
Purpose of data	Project Emission reduction calculation
Additional comment	Fixed for the project for the crediting period. It will be reassessed at each crediting period renewal.

[34] AMS-II.G parameter table “NCVwood”; 20% fuelwood moisture is a conservative estimate for air dried wood: https://uk-air.defra.gov.uk/assets/documents/reports/cat09/1903131256_Seasoning_Wood_Web_Feb_2019_V5.pdf ; IPCC Guidelines for GHG inventories Vol. 2 Ch.2 table 2.9: *“Values were originally based on gross calorific value; they were converted to net calorific value by assuming that net calorific value for dry wood was 20 per cent lower than the gross calorific value.”*

Data/parameter ID	No ID assigned, entirely based on other Parameters
Data/parameter	$EF_{b,useful}$
Unit	tCO_2e per GJ of useful energy
Description	Baseline emissions factor (for entire baseline cooking energy mix)
Source of data	Calculated individually per user, based on other parameters mentioned above. ($EF_{b,useful}$ is included with an own parameter table due to the central position of the parameter).
Value(s) applied	Can vary individually from 0.12 (pure LPG user) to 1.74 (pure charcoal user) Approximated averages per user group: 0.48 for individual users and 0.76 for restaurants.
Choice of data or Measurement methods and procedures	See other parameter tables in this section and calculation in B.6.3, as well as ER calculation sheet
Purpose of data	Emission reduction calculation
Additional comment	Fixed for the project for the crediting period. It will be reassessed at each crediting period renewal. This parameter table was included due to the central position of $EF_{b,useful}$

B.6.3 Ex ante estimation of SDG Impact

SDG 13 (emission reductions)

A step-by-step calculation according to formulas in the MECD methodology is shown in the separate ER calculator in Excel. Below the results of the different steps of the calculations.

Step 1: Baseline emission factor

$$EF_{b,useful} = \sum_k \left(\sum_{i,j} P_{b,i,j} \times Percentage\ of\ fuel\_i \times (EF_{b,i,CO2} \times fNRB_{i,y} + EF_{b,i,non-CO2}) \times NCV_{b,i} \right)_k \div \sum_k \left(\sum_{i,j} P_{b,i,j} \times Percentage\ of\ fuel\_i \times NCV_{b,i} \times \eta_{b,i,j} \right)_k$$

Eq 1 of MECD

Parameter	Average user group		Unit	Explanation	Source
	HH	Restaurants			
$P_{b,charcoal}$	0.250	2.00	t/a	charcoal consumption (t/a)	Individual baseline survey, preliminary average estimates
$P_{b,kerosene}$	0.01	0.01	t/a	kerosene consumption (t/a)	
$P_{b,LPG}$	0.10	0.25	t/a	LPG consumption (t/a)	
$P_{b,wood}$	0.00	0.00	t/a	wood consumption (t/a)	
$Emissions_{b,\ total}$	1.94	13.66	tCO2/a	Baseline emissions total	calculated
EGuseful, total	4.08	18.02	GJ/a	Total baseline energy	
EF_b	**0.48**	**0.76**	**tCO2/GJ**	**Baseline emission factor**	

Step 2: Useful energy delivered by the project

$$\sum_{i,j} P_{b,i,j} \times NCV_{b,i} \times \eta_{b,i,j}$$

$P_{pellets,d,y}$	0.3	0.75	t/a	Pellet consumption per stove	monitored, now preliminary assuming 25kg per month.
$NCV_{pellets}$	0.0157		TJ/T	Net calorific value pellets	laboratory test certificate
$\eta_{pellets}$	54.7%			Mimimoto stove efficiency	ISO test certificate
$EG_{p, useful}$	**2.58**	**6.44**	GJ/a	Useful energy pellets	calculated

Step 3: Baseline emissions

$$BE_y = EG_{p,useful,y} \times EF_{b,useful}$$

$EG_{p, useful}$	2.58	6.44	GJ/a	Useful energy pellets	from above
EF_b	0.48	0.76	tCO2/TJ	Baseline emission factor	from above
BE_y	**1.23**	**4.88**	**tCO2/a**	**Baseline emissions**	calculated

Step 4: Emission reductions

$$ER_y = BE_y - PE_y - LE_y$$

BE_y	1.12	4.88	tCO2/a	Baseline emissions	from above
PE_y	0.05	0.13	tCO2/a	Project emissions	Sum of emissions from electricity, transport and pellet combustion
LE_y	0.00		tCO2/a	Leakage emissions	See B.6.1
ER_y	**1.18**	**4.75**	**tCO2/a**	**Emission reductions per average stove user**	calculated

SDG 3: Avoided or reduced smoke exposure

$$SmokeAvoided_y = N_{improvedstove,y} * Share_{energyreplaced} * Share\_charcoal\_baseline$$

Where:

$SmokeAvoided_y$	Avoided kitchen smoke exposure in person-years
$N_{improvedstove,y}$	Number of smoke-reduced cookstoves (such as portable rocket stoves)
$Share_{energyreplaced}$	Share of pellet energy replacing baseline energy
$Share\_charcoal\_baseline$	Share of charcoal energy usage in the baseline

Preliminary annual result per stove: $SmokeAvoided_y$ = 1 stove * 55% of pellet energy replacing baseline energy * 45% of charcoal usage in the baseline[35] = 0.25 person-years

SDG 3: Reduced health impacts

$Healthimprovement_y = N_{P,y} * Share_{healthimproved,y} *$Where:

$Healthimprovement_y$	Reported health improvements such a cough reduction in person-years
$Share_{healthimproved,y}$	Share of positive respondents (ex-ante estimate). Will be monitored.

Preliminary annual result per stove: $Healthimprovement_y$ = 1 stove * 80%[36] of positive respondents = 0.8 person-years

SDG 5: Time savings
For ex-ante calculations:

$$Timesavedtotal_y = N_{P,y} * Timesavedstove * Share_{female\_cooks,y}$$

Where:

$Timesavedtotal_y$	Accumulated time saved from faster cooking
$Timesavedstove$	Average time saved per stove due to faster cooking (mainly stove lighting)
$Share_{female\_cooks,y}$	Conservative estimate for the share of women cooking. Prelim. 90%

Preliminary annual result per stove: $Timesavedtotal_y$ = 1 stove * 10min[37] * 365days = 61 hours

[35] A preliminary evaluation of the database shows that user households cover 36% of the baseline cooking energy with charcoal, while for restaurants, it is 65%. Considering that 70% of users are households and 30% are restaurants, leads to a weighted average of 45% charcoal usage.
[36] Preliminary estimate: 80% of respondents respond positively
[37] Preliminary estimate: IN average, users will indicate a daily time savings of 10min.

SDG 7: Improved access to clean cooking energy

$$Accesscleanercooking_y = N_{P,y} * Householdsize$$

Where:

$Accesscleanercooking_y$	Access to clean cooking in person-years
$Householdsize$	Average number of persons per household

Preliminary annual result per stove: $Acccesscleanercooking_y$ = 1 stove * 3.9 = 3.9 person-years

SDG 8: Job openings

Calculated based on payrolls. Standardized to person-years of fulltime employment (for example, two persons with half time job are equivalent to one person with a full-time job.

Preliminary annual average: 25 person years (25 full-time employees)

B.6.4 Summary of ex ante estimates of each SDG Impact

SDG 13

Stepwise distribution of an estimated number of 2,500 stoves, 70% to households and 30% to restaurants.

ER measured in tCO2e

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
Year 1	1,161	37	1,124
Year 2	2,323	74	2,249
Year 3	4,645	148	4,497
Year 4	5,807	185	5,662
Year 5	5,807	185	5,662
Total	19,743	629	19,114
Total number of crediting years	5		
Annual average over the crediting period	3,949	126	3,823

SDG 3: Reduced kitchen smoke exposure

Results are based on up to 2,500 households targeted.

Calculated in person-years. Assumption: A household switches from 50% charcoal cooking on a traditional charcoal stove to cooking on a project stove. This household contributes 0.5 person-years of reduced smoke exposure per year.

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
Year 1			124
Year 2			247
Year 3			495
Year 4			618

Year 5			618
Total			2,102
Total number of crediting years	5		
Annual average over the crediting period			420

SDG 3: Reduced reported health impacts of kitchen smoke

Results are based on up to 2,500 households targeted.

Calculated in person-years. Assumption: 80% of households report perceived health improvements. An average household therefore contributes 0.8 person-years of reduced smoke exposure per year.

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
Year 1			400
Year 2			800
Year 3			1,600
Year 4			2,000
Year 5			2,000
Total			6,800
Total number of crediting years	5		
Annual average over the crediting period			1,360

SDG 5: Time saved from faster stove lighting procedure

Results are based on up to 2,500 households targeted. Calculated in total hours saved, assuming 10 minutes saved per day.

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
Year 1			27,375
Year 2			54,750
Year 3			109,500
Year 4			136,875
Year 5			136,875
Total			465,375
Total number of crediting years	5		
Annual average over the crediting period			93,075

SDG 7: Access to clean cooking

Results are based on up to 2,500 households targeted, with average household size of 3.9[38], switching from 45% charcoal usage. (Access to clean cooking is calculated per person, in person-years, accounting thus for 2 persons per stove.

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
Year 1			1,950
Year 2			3,900
Year 3			7,800
Year 4			9,750
Year 5			9,750
Total			31,150
Total number of crediting years	5		
Annual average over the crediting period			6,630

[38] https://housingfinanceafrica.org/documents/2019-kenya-population-and-housing-census-reports/

SDG 8: Job Openings

Calculated in person-years. Preliminarily, 25 full-time employees assumed.

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
Year 1			25
Year 2			25
Year 3			25
Year 4			25
Year 5			25
Total			125
Total number of crediting years	5		
Annual average over the crediting period			25

B.7. Monitoring plan

B.7.1 Data and parameters to be monitored

SDG 13

Data/parameter ID	MECD 9
Data / Parameter	$\eta_{p,d,y}$
Unit	Fraction
Description	Thermal efficiency of the project device
Source of data	Any of the following sources shall be used: - Manufacturer specifications - Third-party certification by a qualified entity - Commercial guarantee Technical reports from the installer
Value(s) applied	Mimi Moto: 54.7% No decrease of this value is considered (see below).
Measurement methods and procedures	Certified ISO test (certificate provided) Certificate of the Advanced Biomass Lab of Colorado State University on non-decrease of efficiency. See A.3.
Monitoring frequency	Annual
QA/QC procedures	Regular maintenance of stoves as described to ensure that ensure full functioning of the critical factors, see A.3.
Purpose of data	This parameter is used in the determination of useful energy
Additional comment	Non-decrease of efficiency is plausible since in case of failures, pyrolysis for gasification and thereby stove operation will stop, see A.3.

Data/parameter ID	MECD 13
Data/parameter	$f_{NRB,y}$
Unit	Fraction non-renewability
Description	Non-renewability status of woody biomass fuel in scenario i during year y
Source of data	Applicable NRB assessment following the main methodology
Value(s) applied	63.9%
Measurement methods and procedures	See fNRB calculation sheet and description under "Determination of the fraction of non-renewable biomass" in Section B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact above for measurement methods and procedures, as well as choice of 63.9% for fNRB. The CDM tool 30 version 4.0 is followed by a lower value is then chosen.
Monitoring frequency	Once at the beginning of a crediting period (see comment below)
QA/QC procedures	Highly conservative choice.
Purpose of data	As applicable, NRB assessment may be used for multiple scenarios where woody biomass is used
Additional comment	It is planned to establish fNRB once at the beginning of a crediting period instead of monitoring it annually. This is conservative given the heavily increasing deforestation in Kenya[39]

[39] Kenya Deforestation Rates & Statistics | GFW (globalforestwatch.org)

Data/parameter ID	MECD 14a
Data / Parameter	$P_{p,d,y}$
Unit	tonnes
Description	The amount of wood pellets used in the project device d in year y, considering cap
Source of data	Direct measurement by pellets sales.
Value(s) applied	300 kg/a per household device (preliminary value, based on 25kg per month) 750 kg/a per restaurant device (preliminary value)
Measurement methods and procedures	Individualized pellet sales data to customers
Monitoring frequency	Continuously, aggregated monthly
QA/QC procedures	Measurement occurs by evaluating sales receipts for packaging units with standardized content. Users pay for pellets; therefore, exhaustive usage is not plausible. Pellets are exclusively distributed by the project and, as described in detail in B.7.3 below, there are no sensible alternative uses for pellets. A corresponding statement is provided separately. Packaging is done with calibrated equipment according to industry standards and relevant national requirements. For calculating ER, a cap will be applied to individual pellet consumption corresponding to a per-capita energy consumption of 0.0045 GJ per day, in accordance with MECD. Where this cap is surpassed, either surplus pellet sales will not be considered, or a plausible explanation will be provided. For example, users may run a small restaurant (this is defined in the user contracts already). In order to avoid crediting of unused pellet at the end of a monitoring period, only transactions taking place after the start of the monitoring period will be considered, and in addition, no ERs will be claimed for the first 5kg used after onboarding of a new user. This has been added under B.7.3.

Purpose of data	Emission reduction calculation
Additional comment	

Data/parameter ID	MECD 15b
Data / Parameter	LE_y
Unit	tCO_2e per year
Description	Leakage in project scenario in year y
Source of data	Option 2 of MECD is followed: Evaluate leakage following the procedure described there.
Value(s) applied	0 (no discount)
Measurement methods and procedures	No significant potential sources of leakage have been identified. The assessment will be repeated at least annually.
Monitoring frequency	Annually
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Emission reduction calculation
Additional comment	Emissions from pellet production are transport are considered as Project Emissions.

Data/parameter ID	No ID assigned, not described in MECD
Data / Parameter	$NCV_{p,i}$
Unit	Terrajoules (TJ)/tonne of fuel
Description	The net calorific value of pellets type i
Source of data	Analyses in independent laboratories
Value(s) applied	0.0157 (value for bagasse pellets, other sources may be used)

Measurement methods and procedures	Tested by "Institute für Bioenergie, Vienna[40].
Monitoring frequency	Annually
QA/QC procedures	Analyses in independent laboratories, cross-check with comparable measurements of other pellets.
Purpose of data	Emission reduction calculation
Additional comment	

Data/parameter ID	No ID assigned, not described in MECD
Data/parameter	$Share_{non\text{-}waste}$
Unit	%
Description	Indicates the percentage of biomass used for pellet production that cannot be deemed waste biomass, i.e., that is regarded non-renewable biomass. For this share, project emissions are calculated as for wood in the baseline.
Source of data	Monitored
Value(s) applied	0% (normally, all pellets are produced from bagasse that is waste biomass).
Measurement methods and procedures	Pellet factory statement
Monitoring frequency	Monthly (monitoring of pellet sources)
QA/QC procedures	N/A
Purpose of data	Emission reduction calculation

[40] Certificate provided to VVB

Non-carbon SDGs

Note: Indicators with standardized units such as “person-years” are used. These units have the advantage of representing absolute (not relative) values, therefore being summable and making projects comparable between each other. (For details, see https://www.fairclimatefund.nl/content/1-onze-aanpak/5-onze-impact/guidance-project-level-sdg-indicators-27mar2021.pdf).

Data / Parameter	SDG 3: Avoided kitchen smoke exposure
Unit	Person-years of cooking with avoided smoke exposure
Description	Accumulated life time of clean cookstove users during which they are able to avoid exposure to smoke from charcoal stoves.
Source of data	Monitoring campaigns
Value(s) applied	Preliminary value: 420 person-years of cooking with reduced smoke exposure, average per year
Measurement methods and procedures	Calculated from the share of total cooking done with the Mimi Moto stove in replacement of charcoal stoves. Only one person assumed to cook per household. Example: A household switches from 45% cooking on a traditional charcoal stove to cooking on the Mimo Moto stove. This household contributes 0.45 person-years of reduced smoke exposure per year.
Monitoring frequency	Annually
QA/QC procedures	Certificate on emissions from MimiMoto stove.
Purpose of data	Calculation of project scenario
Additional comment	The baseline smoke exposure is not measured physically: instead an average smoke exposure from traditional cookstoves is assumed. If possible, accompanying studies directly measuring smoke concentration (e.g., PM, CO, etc.) might be used to substantiate the findings.

Data / Parameter	SDG 3: $Healthimproved_y$: Reduced reported health impacts of kitchen smoke
Unit	Person-years with reported health improvement
Description	Accumulated life time of clean cookstove users during which they report improved health due to avoided exposure to smoke from charcoal stoves.
Source of data	Monitoring campaigns
Value(s) applied	Preliminary value: 1,360 person-years of cooking with reduced smoke exposure, per year, assuming 80% of positive respondents.
Measurement methods and procedures	Calculated from the total number of stove users time the share of respondents indicating health improvements ($Share_{healthimproved,y}$). Only one person assumed to cook per household.
Monitoring frequency	Annually
QA/QC procedures	Certificate on emissions from Mimimoto stove.
Purpose of data	Calculation of project scenario
Additional comment	The parameter is based on subjective user perceptions. If possible, accompanying studies directly measuring smoke concentration (e.g., PM, CO, etc.) might be used to substantiate the findings.

Data / Parameter	SDG 5: Time saved from faster cooking
Unit	Total accumulated hours saved by woman using the Mimimoto stove.
Description	Time savings achieved by women who can reduce cooking times due to faster cooking on the Mimi Moto stove.
Source of data	Stove database, fuel sales showing stove usage.
Value(s) applied	10 minutes per day
Measurement methods and procedures	Calculated based on Mimi Moto stove usage and survey data on time consumption on baseline stoves.
Monitoring frequency	Annually
QA/QC procedures	
Purpose of data	Calculation of project scenario
Additional comment	

Data / Parameter	SDG 7: Access to clean cooking technology
Unit	Person-years of using clean cookstoves
Description	Accumulated life time of persons in households during which the clean Mimimoto stove is used instead of a charcoal stove.
Source of data	Monitoring
Value(s) applied	6,630 (preliminary value)
Measurement methods and procedures	Calculated from the share of total cooking done with the Mimimoto stove in replacement of charcoal stoves. Example: A household of 4 switches from 45% charcoal cooking on a traditional charcoal stove to cooking on the Mimomoto stove. This household contributes 1.8 person-years of clean cooking per year.
Monitoring frequency	Annually
QA/QC procedures	N/A
Purpose of data	Calculation of project scenario
Additional comment	Calculated per household member, since SDG focuses on clean energy access, not smoke reduction.

Data / Parameter	SDG 8: $Jobopenings_y$ Employment generation by the project
Unit	Person-years of employment due to the project
Description	Employment generation standardized to person-years of fulltime employment.
Source of data	Monitoring
Value(s) applied	Preliminary value: 25
Measurement methods and procedures	Calculated based on payrolls. Standardized to person-years of fulltime employment (for example, two persons with half time job are equivalent to one person with a full-time job.
Monitoring frequency	Annually
QA/QC procedures	
Purpose of data	Calculation of project scenario
Additional comment	

B.7.2 Sampling plan

No sampling is applicable, since monitoring of fuel consumption covers 100% of project devices.

The only variables that are not monitored continuously through stove distribution and pellet sales are $Healthimproved_y$, the perceived effects of smoke reduction, and time saved from faster cooking. Since both are rather qualitative variables, non-systematic sampling will be applied. Perceptions of at least 30 users will be gathered, for example, during a defined period before verification, users that get in touch with project staff for fuel purchases or stove maintenance will be asked about perceived health impacts and time savings.
Alternatively, phone interviews will be conducted with at least 30 randomly selected users. This number is deemed sufficient because both are qualitative parameters that does not require high statistical significance.

B.7.3 Other elements of monitoring plan

A. Unique numbering system

Each participating stove user is assigned a unique ID by the system at time of subscription. Moreover, each stove has a unique serial number (usually 9 digits). Unique transaction numbers identify fuel purchases through mobile money service providers such as MPESA.

B. Stove Deployment Record:

The project will maintain and continuously update an electronic database with all stove records. Data will include:

i. Customer ID
ii. Date of stove installation
iii. User name
iv. Address
v. Telephone number (if available)
vi. GPS coordinates of the customers (if available)
vii. Customer indications on baseline fuel consumption

Data entries are created when inserting user data in a smartphone app. Each user also signs a paper contract when onboarding, containing the data saved in the database, except customer ID and GPS data which are automatically assigned by the system when

inserting user data. Paper contracts are stored and serve as a backup in well secured containers on site, with an administrative filing system in place that makes it possible to trace where every stored contract is. Scans of the paper contracts are also available through a link in the database.
A video is provided to the VVB showing the procedure of collecting user data.

C. Fuel sales record

Each pellet purchase transaction is recorded and automatically assigned to the corresponding customer, identified by a unique transaction number provided by the money service provider MPESA. Thus all pellet purchase can be tracked on an individual stove user level.

The pellet packaging process is as approved by the Kenya Bureau of Standards[41]. It is currently done as follows (modifications are possible in the future):

Pellets are received at the warehouse in either bulk super sacks, or 50kg woven gunia sacks.

Each is opened and inspected, then screened by hand over wire mesh trays to filter out fines, short pieces, and inspect for contaminants.

Moisture readings are taken using a moisture analyzer VPB-10 from Henk Maas Weegschalen B.V. Netherlands.

Sacks are filled and weighed individually on scales that have been calibrated, tested, and sealed by the Kenya Bureau of Standards.

Each individual sack contains an authorization mark from KEBS, certifying compliance.

[41] Evidence and description provided to VVB

Measures to guarantee that pellets are exclusively used in project stoves

There are no sensible alternative usages for wood pellets in Kenya than usage in the gasifier stoves distributed by EcoSafi, because:

- There are no other pellet-based cooking solutions being distributed to households in Kenya
- Wood pellets are not suitable as a cooking fuel in other common cookstoves than gasifiers, as they cannot even be lighted properly.
- Other potential usages for wood pellets, such as in biomass boilers or for animal bedding, would make no economic sense at all, due to the extremely high comparative high cost of wood pellets. (The cost of pellets is at least 21KSH per kg, while briquettes are closer to 14KSH, with a comparable calorific value).

The same is confirmed by the independent expert Anneliese Gill-Wiehl, a researcher of Berkeley University who published numerous scientific papers on clean cooking in Nairobi[42].

Moreover, the pellet producer, who also produces briquettes, also confirms the huge cost-advantage of briquettes, as well as the lack of demand for pellets for other purposes[43].

Together with each monitoring event, these points will be evaluated again.

Moreover, it will be shown in each monitoring event that $P_{p,d,y}$ exclusively refers to pellets that are used in stoves included under this project. This will be done by:

- Considering only pellet purchases taking place after the start of the monitoring period, and in addition, not claiming ER for the first 5kg used after onboarding of a new user.
- Investigating any individual user's pellet consumption that surpasses 480kg per year, below the MECD cap of 0.0045GJ per capita and day, in order to either find a transparent and plausible explanation (the user may run a restaurant which is noted in the user contracts, household may be very large, the user may order

[42] Evidence provided to VVB. See, for example: https://erg.berkeley.edu/people/annelise-gill-wiehl/ https://www.nature.com/articles/s41560-022-01126-2

[43] Confirmation provided to VVB, see also www.vumabiofuels.com

pellets for his neighbor also), or to consider only the amount below the cap for ER calculation.

D. Data preparation for verifications

Stove and fuel sales are recorded continuously in real time. For verifications (and regular inventories), data are exported from the data management system to Excel. The resulting spreadsheets will contain a complete list of users, together with pellet purchase data individually assigned to these users (individual users and resellers). On a separate sheet, pellet purchase per customer will be disaggregated by single purchase events, specifying purchase date and packaging sizes.

The key monitoring parameter $P_{p,d,y}$ (amount of wood pellets used in the project devices in year y) will be derived by summarizing pellet purchases by all individual users in the corresponding time

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1 Start date of project

16/06/2021

(The date when the roll-out of stove distribution started).

C.1.2 Expected operational lifetime of project

10 years: The project's lifetime is longer than the clean cookstove's lifetime because during the project's lifetime, clean cookstoves can be replaced. Moreover, as explained under A.3, the stove lifetime can be longer than 5 years due to constant maintenance and exchange of spare parts.

C.2. Crediting period of project

C.2.1 Start date of crediting period

16/06/2021

C.2.2 Total length of crediting period

5 years (renewable)

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1 Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in Appendix 1, ongoing monitoring is summarised below.

PRINCIPLES	MITIGATION MEASURES ADDED TO THE MONITORING PLAN
N/A	N/A

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project reflects the key gender issues and requirements of gender-sensitive design and implementation of the project. SDG#5 is one of the impact areas of the project. The introduction of new and clean cookstoves will result in reducing use of charcoal at the user's place. In Kenya, cooking activity at household level is mostly still managed by women. Therefore, women are more exposed to the indoor air pollution and the associated hazard and will benefit more from clean cookstoves. Time savings due to faster cooking also provides special benefits to women.

Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The National Policy on Gender and Development in Kenya[44] defines goals and measures towards a just, fair and transformed society free from gender-based discrimination in all spheres of life practices. The project aligns with this policy by providing practical benefits to women who use the clean stoves (see above), but also by directly dealing with women as primary and fully responsible customers and workers in the large majority of cases, thereby fully respecting women's equal rights and economic independence.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	All the gender safeguarding principles and requirements are carefully assessed and an expert is not required.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	The project team responsible for organizing the stakeholder consultation did ensure the gender representation at the meeting. So, no expert is required.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1 Summary of stakeholder mitigation measures

No mitigation measures were identified at the local stakeholder meeting.

[44] psyg.go.ke/wp-content/uploads/2019/12/NATIONAL-POLICY-ON-GENDER-AND-DEVELOPMENT.pdf

E.2 Final continuous input / grievance mechanism

METHOD	INCLUDE ALL DETAILS OF CHOSEN METHOD (S) SO THAT THEY MAY BE UNDERSTOOD AND, WHERE RELEVANT, USED BY READERS.
Continuous Input / Grievance Expression Process Book (mandatory)	EcoSafi store on Moi Avenue opposite Timani Primary School and First Love Penacostal Church
GS Contact (mandatory)	help@goldstandard.org
Telephone access (optional)	Tom Price CEO EcoSafi Cellphone: +254 0796 549 100 Ms Corcoran +254 0704 963 276
Internet/email access (optional)	Mr. Tom Price CEO EcoSafi tom@bettercookingcompany.com Ms Corcoran info@ecosafi.com
The Nominated Independent Mediator (optional)	N/A

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into SECTION D above. Please refer to the instructions in the Guide to Completing this Form.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in	No	1. The project does not negatively affect human rights. Participation is fully voluntary. 2. No discrimination occurs because the project offers clean cookstoves to interested people who can freely decide to use them or not. The project is basically targeting women in poor households depending on charcoal for cooking.	n.a.

the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion			
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination,	No	1. The project focuses on a gender sensitive design and planning. Women were involved from the beginning when exploring the feasibility of the project and designing its implementation. Furthermore, the Stakeholder Consultation was organized with a majority of participants being women. 2. The project does apply the principles of non-discrimination and equal treatment in distribution and usage of the products. It embraces the spirit of the Kenyan Policy on Gender and Development[45], the national Labour Laws[46] and ILO guidelines. EcoSafi also ensures equal pay for equal work during any activity in the project. 3. The project applies the principles of the Kenyan Policy on Gender and Development (see section D.2). 4. n.a.	n.a.

[45] psyg.go.ke/wp-content/uploads/2019/12/NATIONAL-POLICY-ON-GENDER-AND-DEVELOPMENT.pdf
[46] www.ilo.org/ifpdial/information-resources/national-labour-law-profiles/WCMS_158910/lang--en/index.htm

equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)			
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers	No	There are no health risks associated to the usage of the clean cookstoves offered. To the contrary, the reduction of kitchen smoke is an important objective of the programme and part of monitoring under SDG 3. Due to the principle of pyrolysis producing wood gas, the stoves burn as clear as LPG stoves.	n.a.

and the community			
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The Project Area consists of the compounds of the participating households and institutional users. Therefore, cultural heritage sites are generally not included in the Project Area.	n.a.
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	To the contrary, the provision of clean cookstoves to households improves their economic situation and rather mitigates their relocation for economic reasons.	n.a.
>>			
Principle 4.3 Land Tenure and Other Rights			
Does the Project require any change, or have any uncertainties related to land tenure	No	Land tenure and land rights are not relevant for or affected by the program at all.	n.a.

arrangements and/or access rights, usage rights or land ownership?			
For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership? Examples include, but are not limited to water access rights, community-based property rights and customary rights.	No	Land tenure and land rights are not relevant for or affected by the program at all.	n.a.
Principle 4.4 – Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	The project is in urban areas where no land is claimed by indigenous people	n.a.
Principle 5. Corruption			

1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	The program and its associated participants are not and will not be involved in any kind of corruption, in accordance with Kenya's Anti-Corruption Policy (www.pc.go.ke/sites/default/files/ANTI-CORRUPTION%20POLICY-JUNE-2010.pdf)	n.a.
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations	No	1. The project will respect Kenyan Labour regulations and ILO standards. It will meet the highest standards according to: 2. https://mywage.org/kenya/documents/120606-decentworkcheck-kenya_NEW.pdfEcosafi workers will not be hindered in establishing or joining labour organizations. 3. Proper labour contracts will be signed with all workers by Ecosafi, including at least the following element: a. Working hours not exceeding 48 hours per week b. Description of duties and tasks c. Remuneration (including overtime provisions) d. Modalities on health insurance e. Termination modalities for both sides f. Annual leave >10 days per year 4. All workers for EcoSafi shall be over 18. Age will be checked using national identity cards 5. Safety measures will be followed, in accordance with the Occupational Safety and Health Act 2007. Any accidents or incidents during work for EcoSafi will be duly reported and documented. Appropriate response measures will be discussed and implemented.	n.a.

3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary			

resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and			

incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	No	The program has nothing but positive economic impacts on the local economy, creating jobs and enabling women to reduce fuelwood expenses.	n.a.
>>			
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	As per the application of the GS methodologies no increase of emissions over the baseline scenario occurs.	n.a.
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a	No	Since the project stoves rely on residual biomass (sawdust), household energy sourcing from local resources will be reduced. Therefore, the energy supply to other users is not negatively affected, if anything it is positively affected.	n.a.

national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?			
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project has no direct impact on natural water resources. If anything, there is a positive impact through the reduction of charcoal production related to deforestation and desertification.	n.a.
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a) Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the	No	a) The project has no direct impact on natural water resources. If anything, there is a positive impact through the reduction of charcoal production related to deforestation and desertification. b) There is no susceptibility to erosion or water body instability.	n.a.

natural pattern of erosion? b) Is the Project's area of influence susceptible to excessive erosion and/or water body instability?			
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The project does not involve the use of land and soil.	n.a.
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	If any, the project will have a positive impact by mitigating climate change and reducing charcoal production related to deforestation and desertification.	n.a.
>>			

Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	no	There is no possible impact of GMOs on the use of clean cookstoves.	n.a.
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	no	There is no use of pollutants in the program. Cookstoves and batteries will be taken back by the project after usage.	n.a.
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-	no	No chemicals are used by the project.	n.a.

hazardous chemicals and/or materials?			
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	no	This project will not use any kind of pesticide or herbicides or chemical fertilizer.	n.a.
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	no	The project does involve the harvesting of forests, but: Since the project will reduce the consumption of charcoal significantly, harvesting of forests will be reduced, and hence the sustainable management of forests will be fostered and there will be a positive impact on biodiversity and ecosystem functionality. The fuel used by the project is made from residual biomass.	n.a.
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	no	The impact of the program is only on cooking energy, but not on the nutritional quality or quantity of food available to households. Although, it is possible that the savings on expenditures for fuel wood will be used by some families to increase or improve their food supply.	n.a.
>>			
Principle 9.9 Animal husbandry			

Will the Project involve animal husbandry? >>	no	This project will not involve any animal husbandry.	n.a.
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified? >>	no	The program promotes interventions only at the household level, not affecting critical ecosystems.	n.a.
Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? AND/OR	no	The program promotes interventions only at the household level, not affecting endangered species.	n.a.

Does the Project potentially impact other areas where endangered species may be present through transboundary affects?			
>>			

APPENDIX 2 - CONTACT INFORMATION OF PROJECT DEVELOPER(S)

Organization name	Better Cooking Company Limited
Registration number with relevant authority	Kenyan company registration number PVT-EYUJPVK
Street/P.O. Box	PO 1908 Karen, Nairobi
Building	n.a.
City	Nairobi
State/Region	n.a.
Postcode	n.a.
Country	Kenya
Telephone	+1 801 712 5371
E-mail	info@ecosafi.com
Website	Ecosafi.com
Contact person	Tom Price
Title	CEO
Salutation	Mr.
Last name	Price
Middle name	Watson
First name	Tom
Department	n.a.
Mobile	+1 801 712 5371
Direct tel.	+1 801 712 5371
Personal e-mail	tom@bettercookingcompany.com

APPENDIX 3 - LUF ADDITIONAL INFORMATION

Risk of change to the Project Area during Project Certification Period:	
Risk of change to the Project activities during Project Certification Period:	
Land-use history and current status of Project Area:	
Socio-Economic history:	
Forest management applied (past and future)	
Forest characteristics (including main tree species planted)	
Main social impacts (risks and benefits)	
Main environmental impacts (risks and benefits)	
Financial structure	
Infrastructure (roads/houses etc):	
Water bodies:	
Sites with special significance for indigenous people and local communities - resulting from the Stakeholder Consultation:	
Where indigenous people and local communities are situated:	
Where indigenous people and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance:	

APPENDIX 4 - DESIGN CHANGES

A4.1. Details of proposed or actual design change

>>

A4.2. Describe the impacts of design change on the following

a. Additionality

>>

b. Applicability of methodology and other methodological regulatory documents with which the project activity has been certified

>>

c. Compliance with the monitoring plan of the applied methodology

>>

d. Level of accuracy and completeness in the monitoring of the project activity compared with the requirements contained in the registered monitoring plan

>>

e. Scale of the project activity

>>

f. Stakeholder consultation

>>

g. Sustainable development criteria

>>

h. Safeguarding assessment

>>

i. Compliance with applicable legislation

>>

j. <u>Only for LUF Projects</u>: Transparent summary of all approved changes in Project Area, Eligible Area and accompanying changes in ex-ante emissions removals.

DATE OF APPROVED DESIGN CHANGE (MM/DD/YYYY)	PROJECT AREA (HA)		ELIGIBLE AREA (HA)		EX-ANTE ESTIMATE (TCO2E)	
	INCREASE OR DECREASE ?	VALUE (HA)	INCREASE OR DECREASE?	VALUE (HA)	INCREASE OR DECREASE ?	PERCENTAGE (%)

Revision History

Version	Date	Remarks
1.5	29 June 2023	Editorial changes to match V2.1 of the Safeguarding Principles Requirements
1.4	21 June 2023	Editorial changes to match V2.0 of the Safeguarding Principles Requirements
1.3	14 April 2023	Integrated the design change memo as annex of the document. Editorial changes
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption